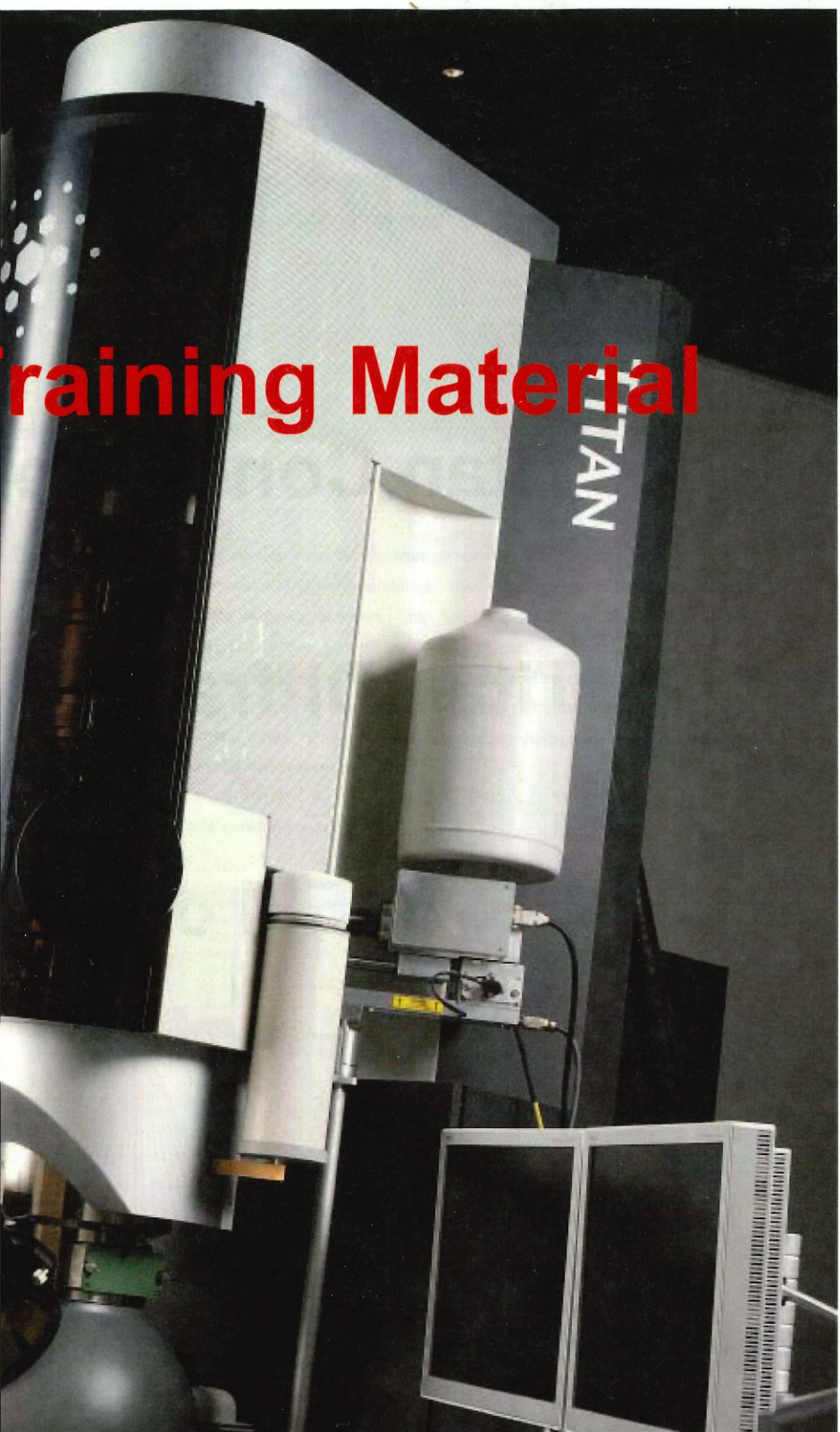

raining Material
TAN

Contents:

1. Titan Condenser

2. Titan Diffraction

3. HR-STEM on Titan

FEI COMPANY™
TOOLS FOR NANOTECH

Module 1
Titan Condenser System

By Dong Tang and Peter Tiemeijer

The complete condenser system of Titan for illumination consists of 6 lenses: gun lens, the first condenser, the second condenser, the third condenser, mini-condenser and upper-polepiece of objective lens. Different from its previous series, Titan has an extra condenser 3, which makes illumination more flexible. Condenser system is controlled mainly through Beam Settings, Aperture and FEG Register control panels in Titan user interface, as well as beam focus knob, spot size button, gun lens and condenser stigmator. The purpose of this module is to highlight the major differences from the Tecnai condenser system and to help a user understand the uniqueness of the Titan condenser system.

1. Condenser system and condenser zoom

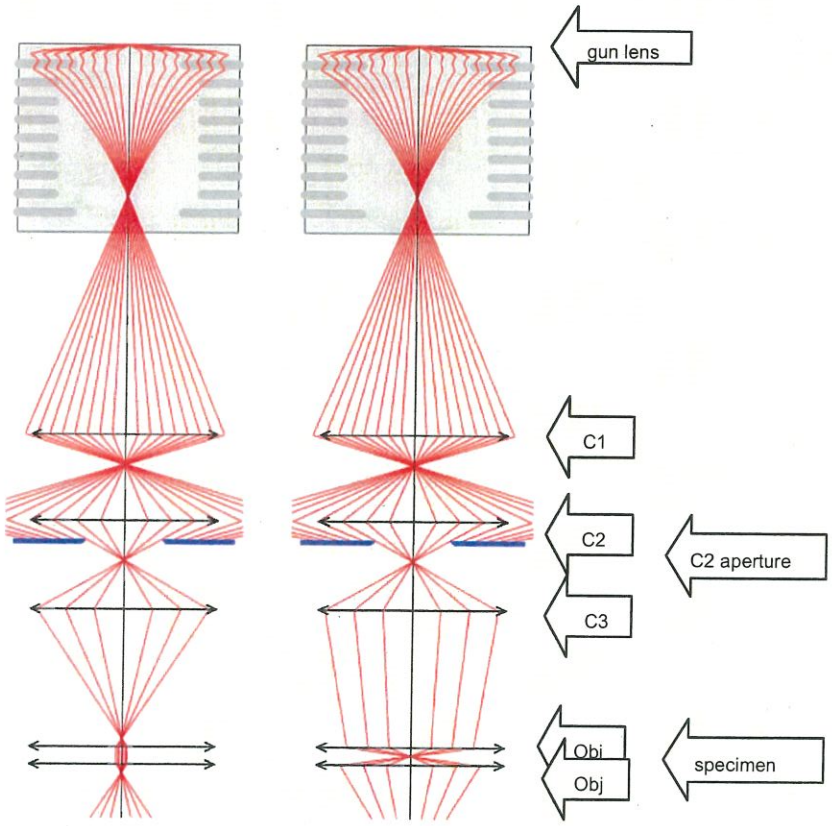


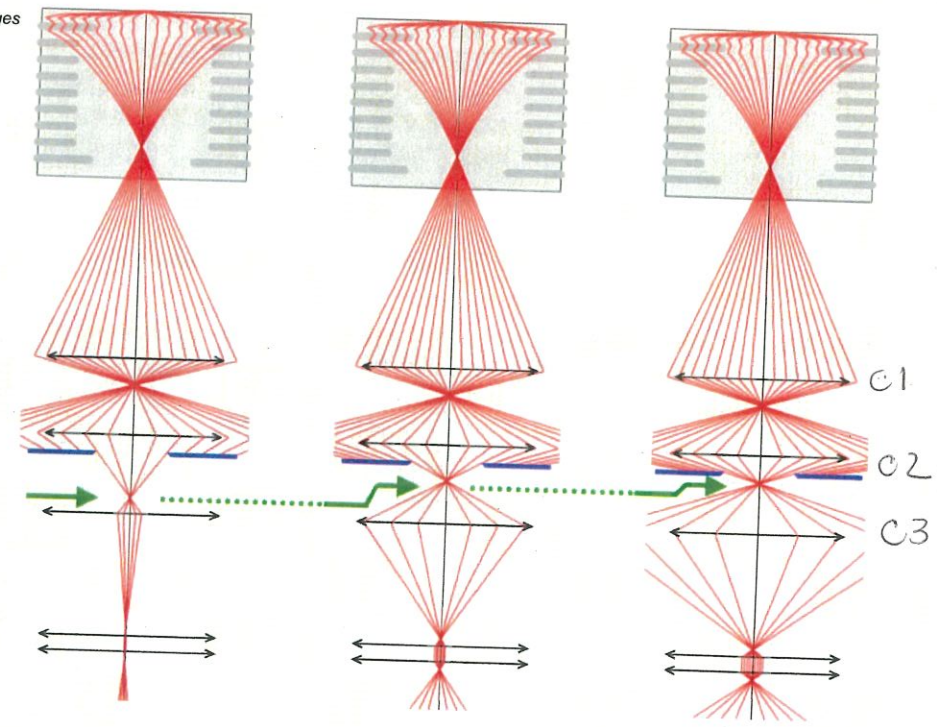
Figure 1: A simplified overview of Titan condenser system in TEM mode (left) and Probe mode (right).

Figure 1 shows the positions and functionalities of these lenses in the condenser system. For simplicity, mini-condenser is not shown. It will be introduced later in this module. Gun lens, an electrostatic lens, is located immediately below the extractor. It sets the first beam crossover that

is in turn the object of the first condenser lens. The last beam crossover from the condenser system is set either at the front focal plane of the objective lens upper-polepiece (TEM mode and parallel illumination) or at the specimen position (Probe mode and convergence illumination).

If we can keep the first and last beam crossover position unchanged while altering the lens settings, we then have a very flexible illumination. This is actually called condenser zoom. For example, by changing the position of an intermediate beam crossover between C2 and C3 in TEM mode, we can control the diameter of the illuminated area, keeping other conditions unchanged such as parallel illumination. This is schematically showed in figure 2. To do this area zooming while keeping the illumination parallel, we need change C2 and C3 simultaneously. This is coordinated in Titan software while you turn the intensity knob.

Figure 2: A zoom function that changes size of parallel illuminated area.



In a two-lens condenser system, the truly parallel illumination is only at a unique C2 setting. Away from this setting, either overfocus or underfocus, the illumination starts to diverge or converge. In a two-lens system, therefore, an image is acquired not with the exactly parallel illumination, or in a condition that the illumination at the edge is more inclined than the centre.

The advantages to keep a true parallel illumination in TEM mode are obvious. First of all, the theories of image formation and, then, image interpretation are based on a perfect parallel illumination. For example, the inclined illumination may introduce extra aberrations, e.g. astigmatism, defocus and coma etc, to the image. Secondly, nano-beam diffraction, a technique to obtain nano-range diffraction information using a small focused beam with a size of the same order, needs the illumination as parallel as possible so that to avoid the overlap of adjacent diffraction discs. Another application is in TEM tomography, where a specimen might be tilted to a very high angle. If the illumination is not really parallel, the magnification may vary from one end to another at the high specimen tilting angle.

The same kind zooming (synchronization of C2 and C3) applies to Probe mode, too. It changes convergence angle rather than the size of illuminated area. In other words, the illumination probe is always focused on the specimen while the beam convergence angle may be varied. One of the main applications of convergence angle zoom is high-resolution STEM, where the condenser aperture size relatively to the beam convergence angle (seen in figure 3 as the dark ring in Ronchigram) is crucial. A too large aperture may introduce artefacts in the image due to the probe tails; a too small aperture will make the illumination dimmer and produce images with low signal-to-noise ratio. Another application is convergence beam electron diffraction, where convergence angle relates to the field-of-view.

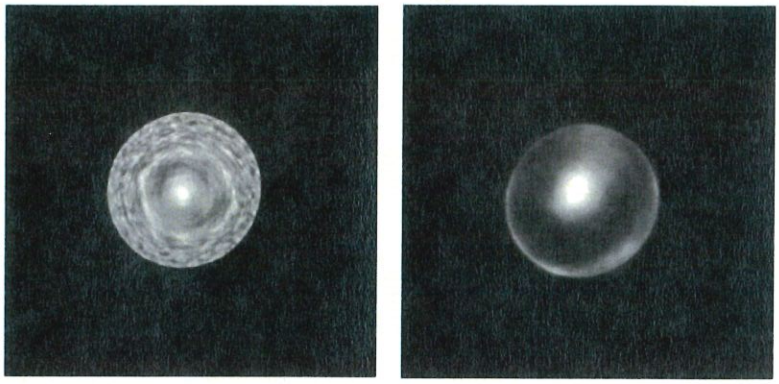
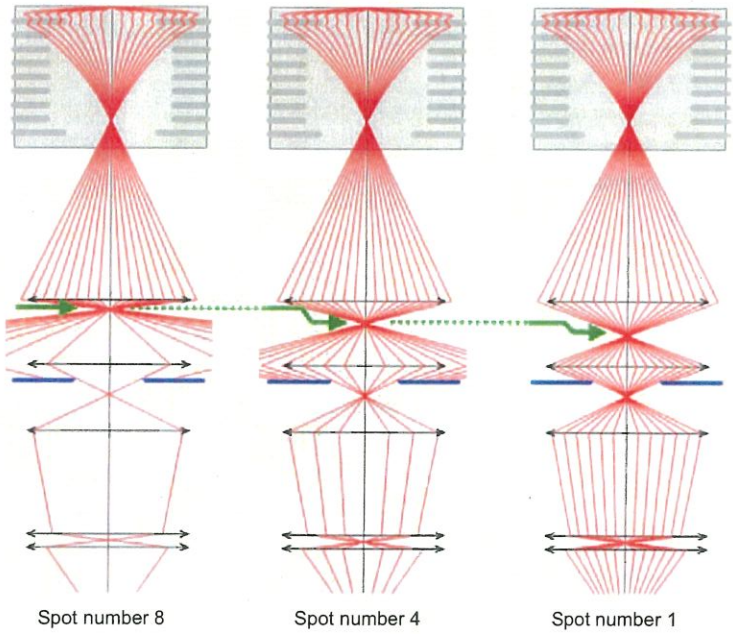


Figure 3: Ronchigrams that show different convergence angles in STEM.

Figure 4: Intensity zoom in Probe mode.



Apart from the above C2-C3 synchronization, we can also change the position of the beam crossover further up, between C1 and C2. This changes the total beam intensity and the minimum focused probe size, typically called intensity zoom. It is outlined in figure 4 for Probe mode. On Titan, we may select 1 out of 11 different spot numbers. The higher number means the less electrons going through the C2 aperture, so that the total beam intensity is lower. Increasing spot number by 1 leads to a beam current reduction of roughly 50%. At the same time, the

magnification power of C2 is lower, because the cross-over before C2 (also an object of the lens) is further away while the cross-over below C2 (also the image of C2) keeps the same position. Therefore, the minimum focused spot size on the specimen is smaller. In Probe mode this leads to better STEM or microanalysis spatial resolution with lower beam intensity. When this kind of zoom is applied to TEM mode, we will have the same effect although normally we do not need beam to be focused to a small spot. Again, software looks after the synchronization of C1 and C2 in intensity zoom when another spot number is selected.

In summary, three condenser system provide us some extra flexibilities and usabilities. There are three kinds of condenser zoom: area zoom, convergence zoom and intensity zoom. Titan software takes care of the syncronization of lenses for the zoomings. To make the zooming system working, some initial calibrations and alignment need to be done (see below).

2. Microprobe and Nanoprobe

The words Microprobe and Nanoprobe were introduced when Twin concept objective lens was implemented in Philips microscopes. Microprobe basically means more parallel illumination, while Nanoprobe refers to convergent illumination. The difference is the setting of mini-condenser, a small lens just above the objective lens. The lens is either optically on or off in Microprobe or Nanoprobe, while physically the same amount of current in the lens coil but in opposite directions. It is to minimise the thermal instability, typically the specimen drift after mode change. In Titan the concepts of parallel and convergent illumination is replaced by TEM and Probe mode. However, it is still possible to select Microprobe and Nanoprobe in the same way as before.

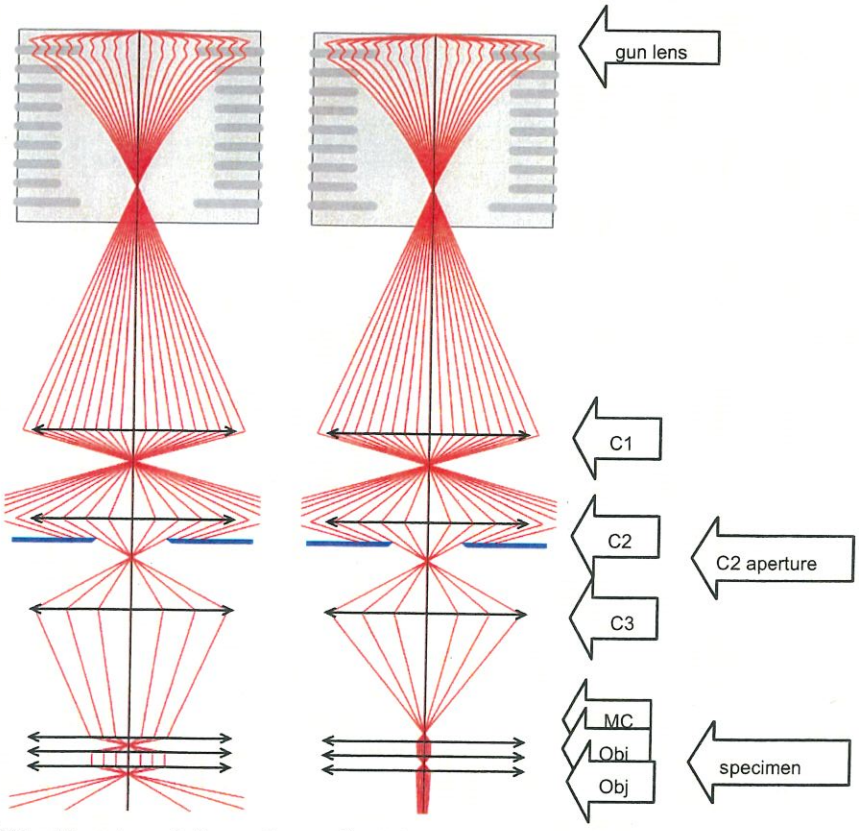


Figure 5: Titan condenser system with mini-condenser on in TEM mode (left) and in Probe mode (right). Comparing to figure 1, where mini-condenser can be understood in off mode, the illumination area is larger in TEM and the convergence angle is smaller in Probe mode.

With mini-condenser in the ray diagram, figure 1 will be slightly modified as figure 5. Although the ray diagram looks more complicated, the final beam crossover is still sitting at the front focal

plane of the upper-polepiece in TEM mode or the specimen position in Probe mode. By default TEM mode is coupled with Microprobe and Probe mode with Nanoprobe, even it is possible to select a different one in some special applications. The merit of Microprobe in TEM is that the maximum area of parallel illumination is 5 times larger. On the other hand, the combination of TEM and Nanoprobe makes the minimum area of parallel illumination 5 times smaller. In probe mode, Nanoprobe produces a maximum convergence angle 5 times larger than Microprobe. Mini-condenser brings more flexibility to the condenser system.

The further discussions of Microprobe and Nanoprobe, on how to obtain the most parallel illumination with different sizes, can be found in the next section.

3. TEM mode and Probe mode

Beam Settings control panel looks very different from Tecnai series (figure 6). As we mentioned before, TEM mode means the final beam crossover is at the front focal plane of the objective lens upper-polepiece, so that to produce a parallel illumination on the specimen. When working in TEM mode, user interface displays the range of parallel illumination and the current size of illuminated area. It should be noted that the area size is linked to the size of C2 aperture used. When a 100μm aperture replaces a 50μm aperture, the illuminated area is doubled. The intensity knob changes the size of the current illuminated area and, at the same time, the Area displayed in the control panel. Because of the limitation of lens strength, when going below the parallel illumination range, the system switches to condensing mode. If going above the range, the system is in spreading mode. In both modes, the beam crossover is not exactly at the front focal plane of the upper-polepiece, but above (condensing) or below (spreading) the plane thus making the beam slightly converging or diverging towards the sample.

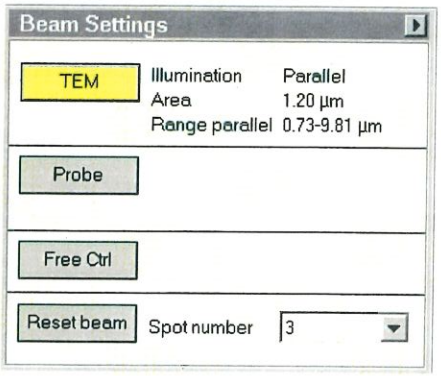


Figure 6: The Beam Settings control panel in TEM mode.

By default, TEM is in Microprobe or mini-condenser on. However, it does not say you cannot select Nanoprobe in Tune flap-out page of Beam Setting (figure 7). This will switch mini-condenser off. Both the minimum and the maximum area that can be illuminated with a parallel beam are five times smaller than in Microprobe. Selecting Auto Zoom button on the same page will keep the illumination size on the view screen (but not the illuminated area on specimen) the same when changing magnification. Or it automatically rescales the illuminated size when magnification changes. When 'MF-Y Fine focus back-focal plane' is selected, multi-function Y is used to fine tune the last beam crossover of condenser system at the front focal plane of the upper-polepiece. This is to guarantee a parallel illumination and to avoid slightly converging or diverging. It is judged by the focused diffraction spots at the back focal plane of the objective

lens. In normal operation it rarely needs to be done. Finally, Reset beam button in Beam Settings panel sets user beam-shift to zero, e.g. setting the left tracking ball to zero.

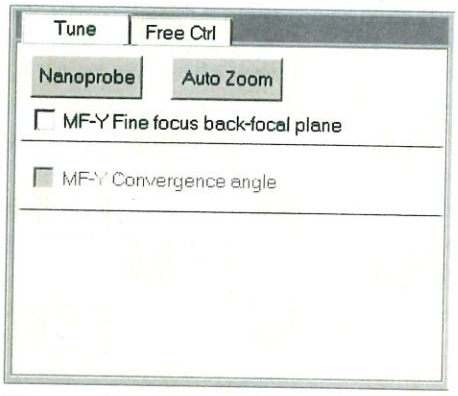
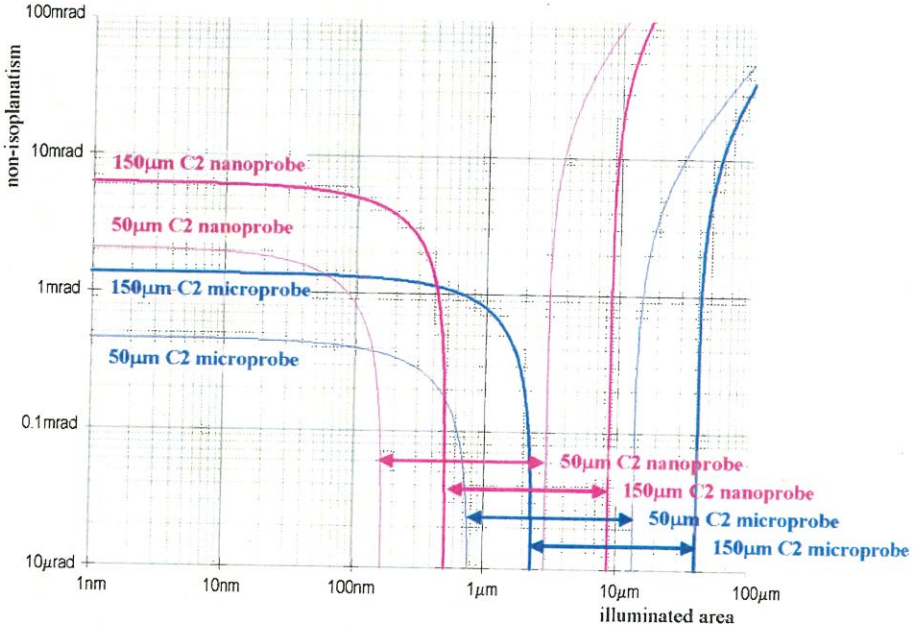


Figure 7: The flap-out Tune page of the Beam Settings control panel in TEM mode.

Beyond the ranges of parallel illumination, the illumination is not truly parallel in either condensing or spreading mode. Figure 8 shows the theoretical relationship of the illuminated area and the beam inclining (convergence or divergence) angle in Microprobe/Nanoprobe and in 50μm/150μm condenser aperture.

Figure 8: Theoretical relationship between the size of illuminated area and the inclining angle (convergence when the area is smaller and divergence when it is larger).



For large area parallel illumination, say from 800nm above, Microprobe should be used and larger condenser aperture will also make the area larger. The applications include BF/DF imaging, diffraction with/without selected area diffraction and TEM tomography etc. If the illuminated area is between 200nm and 800nm Nanoprobe is better, although it seems not necessary for TEM imaging since the convergent angle of Microprobe is in the order of 1mrad or less. The most possible application is then μ-diffraction (spot diffraction with sub-micron probe without using SA aperture). For the illumination area smaller than 200nm, it is better to come back to Microprobe again. This is because the fact that convergence angle changes 5 times

faster in Nanoprobe than in Microprobe. The most important applications are high-resolution TEM and nano-beam diffraction (similar to μ-diffraction but with nano-meter size probe).

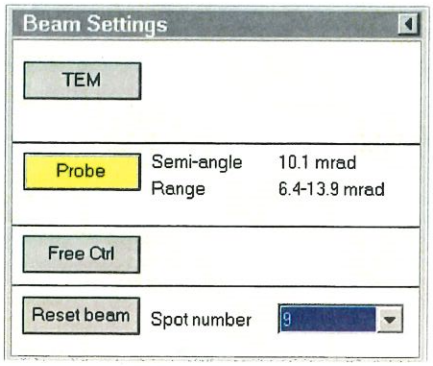


Figure 9: The Beam Settings control panel in Probe mode.

Probe mode produces a focused probe on the specimen for, e.g., STEM, EDX, EELS and CBED. By default it is in Nanoprobe mode for higher beam current and/or large convergence angle. The half-convergence angle of the probe as well as the range of convergence angles are indicated on the right of the Probe button (figure 9). The focus of the probe is controlled by the Intensity knob on the left hand control panel, except in STEM mode where it is usually controlled with the Focus knob on the right hand control panel. The default half-convergence of 10 mrad, together with a 50μm C2 aperture, is an optimal value for high-resolution STEM on Titan without probe Cs-corrector. The half-convergence angle can be varied in the flap-out Tune page by selecting 'MF-Y Convergence angle' (figure 10).

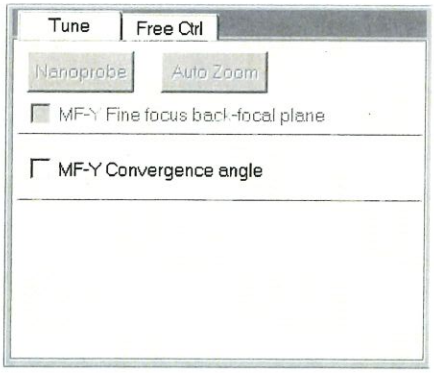


Figure 10: The flap-out Tune page of the Beam Settings control panel in Probe mode.

As the illuminated area size in TEM mode, the half-convergence angle and the range are directly proportional to the size of the C2 aperture. A 100μm aperture will double the convergence angle of the one using a 50μm aperture, as well as the maximal angle in the range. The convergence zoom is in fact the synchronization of C2 and C3 and it is schematically showed in figure 11. In probe mode, the fine tune of the beam crossover at the front focal plane of the upper-polepiece is not necessary, because it now located at the specimen. This is no different from focusing beam to a smallest spot.

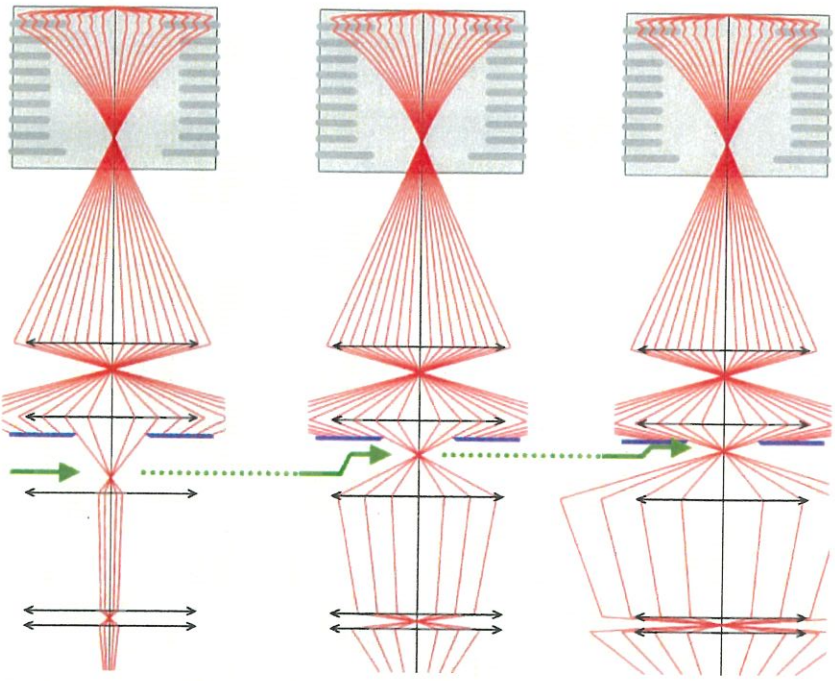


Figure 11: Convergence zoom in probe mode.

4. Free control

The third mode of beam settings is free control. In this mode, a user can set up a condition of condenser system which is not covered in TEM and Probe modes. When it is entered, the Free Control flap-out page automatically pops up (figure 12), where one can set up the condenser system.

When coming from TEM or Probe to Free Control mode, the illumination keeps the same. But now several additional control options become available in the free control page. When switching back to TEM or Probe mode, the illumination is set to fit whatever the restrictions in these two modes. Most of the options in the free control page are clear. We only emphasize here C2/C3 off mode and Normal/Large angle range.

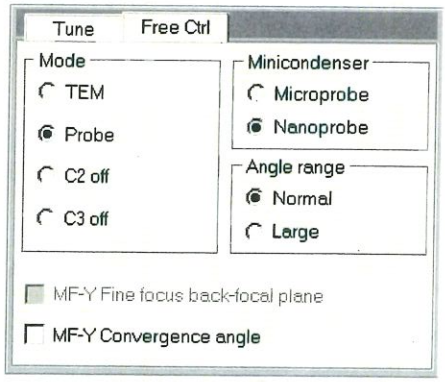
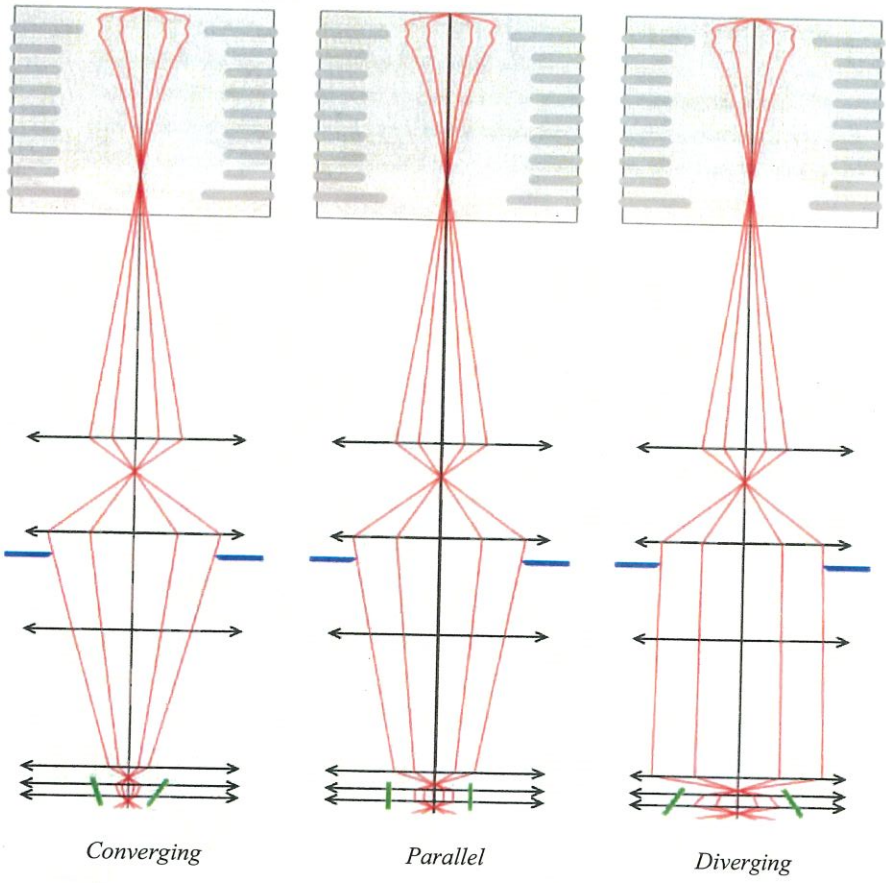


Figure 12: Condenser system free control page.

In the modes of C2 off or C3 off, the second or third condenser lens is set to zero and it behaves as a two-condenser system, similar to a Tecnai or other microscopes. The condenser zooming system is then disabled. Although normally these two modes are not often used, they may be

beneficial to, for example, the circumstances that the condenser lenses or C2 – C3 zoom have become misaligned for some reasons. In C2 off mode, the Intensity knob controls condenser 3 and the C3 aperture is the beam performing aperture. In C3 off mode, the Intensity knob controls condenser 2 while the C2 aperture is the beam limiting aperture. It must be clear that the beam is generally no longer truly parallel except at a particular C2/C3 setting, as illustrated in figure 13.

Figure 13: In C3 off mode, the condenser system is similar to that in Tecnai. The beam changes from converging to parallel then to diverging when beam-focusing. However, the deviation from a perfect parallelism is not as much as that showing in the figure.



Let us look back to figure 11 for convergence angle zoom. When both condensers are strongly excited, we expect to see a real beam crossover between C2 and C3. This intermediate crossover between C2 and C3 can also be a virtual one. The former option gives more optical flexibilities, resulting in a larger range of convergence angles than the latter. Accordingly the first option is called large range while the latter option is called normal range (figure 14).

Because of its convergence angle range, the large range option is especially suitable for LACBED (large angle convergent beam electron diffraction). However, the strongly excited C2 and C3 lenses in the option also contribute strongly spherical aberrations to the probe performing, especially at small convergence angles. Therefore, in order to obtain a probe with low lens aberration, normal range option is a preferred mode, such as in the applications like STEM and EDX where aberrations determine spatial resolution. The spherical aberration of a weakly exited C2 lens in normal range option contributes little to the total probe aberrations.

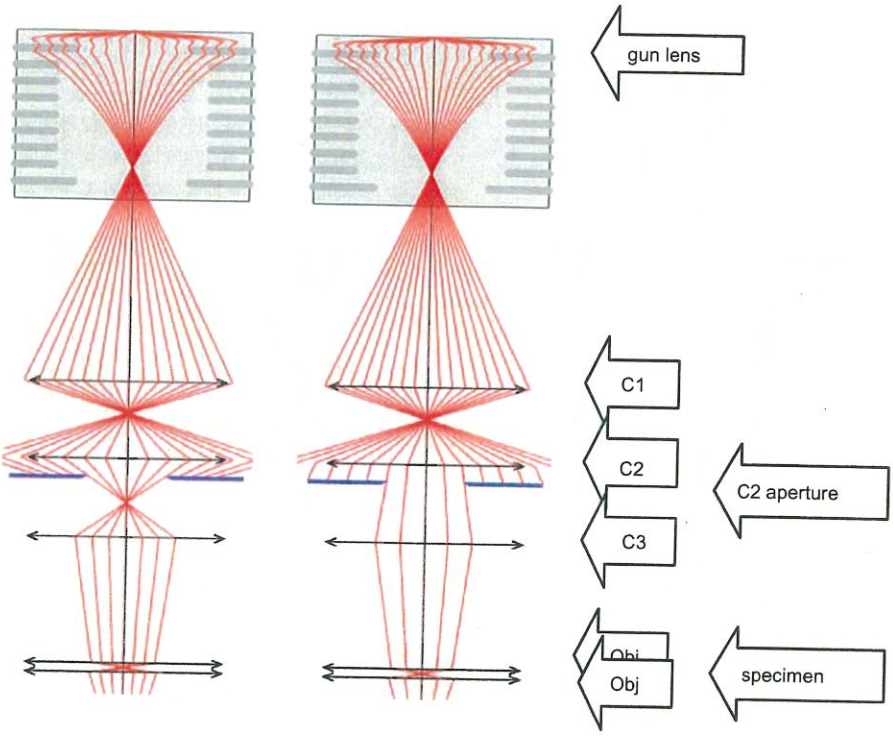


Figure 14: In large range option there is a real beam crossover between C2 and C3 (left). In normal range, the beam crossover is a virtual image (right).

5. Alignment

The alignment principle of Titan from version 1.0 on is that the administrator is responsible for the full column alignment, while users only do the adequate and, mostly, direct alignments. The rule applies to condenser system alignment, too. Only administrator (plus service and factory, of course) users are allowed to do full condenser alignment. For a normal user, the only available alignment folder is Gun.

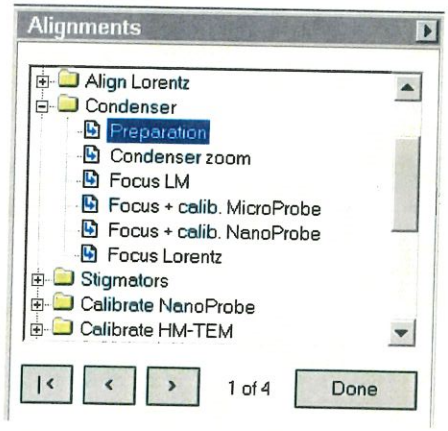


Figure 15: Condenser alignment folder for an administrator user.

If you are an administrator, you may find the following alignment procedures in Condenser alignment folder (figure 15):

1. **Preparation**: is to calibrate the strengths of the condenser lenses (C1, C2 and C3) and find the precise position of the first beam crossover (gun crossover) before C1. The way to do it is to have two lenses strongly excited while calibrating the third.

2. **Condenser zoom**: is to minimise the beam movement in area zoom and convergence zoom. It actually aligns C3 to C2.
3. **Focus LM**: is to locate the final beam crossover position in Probe and TEM, respectively, in microscope LM mode. The criterion is a focused beam in Probe or a focused diffraction pattern in TEM.
4. **Focus + calibration Microprobe**: is doing the same as above but in high-magnification range. Besides, the size of the illuminated area is calibrated. A cross-grating standard sample is required for this calibration.
5. **Focus + calibration Nanoprobe**: is again a similar step but in Nanoprobe mode. The convergence angle is calibrated based on the diffraction rings of poly-crystalline gold.
6. **Focus Lorentz**: does the same alignment in Lorentz mode, if a Lorentz lens is implemented.

One does not need to fully align the condenser system frequently once it has been aligned by going through the Condenser alignment folder. The strengths of condensers are not variable, neither the final beam crossover positions, as long as the specimen is at eucentric height. The critical alignment of the zooming system becomes to find the position of the first beam crossover, or gun crossover. In the full alignment folder, it is achieved in the first step of preparation. However, gun crossover is the function of extraction voltage, gun lens, high tension and gun tilt/shift. Whenever those are substantially changed, a re-alignment of gun crossover is needed. For this reason, gun crossover alignment procedure is also found in Gun alignment folder that is the only accessible folder by non-administrator users (figure 16).

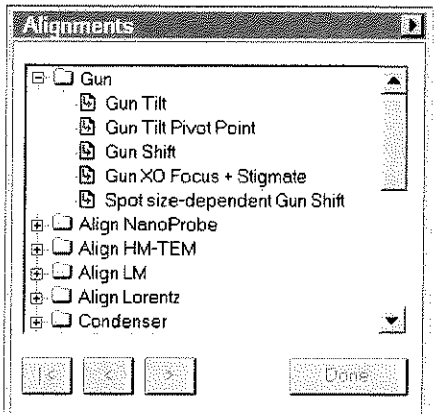


Figure 16: Gun alignment folder for administrator.

When a new and substantially different gun setup is used, it is advised to do all the alignments in gun folder, e.g. aligning gun as well as gun crossover. The Gun XO Focus + Stigmator folder does not only locate the new gun crossover position in height, but also correct condenser astigmatism using the gun stigmators. The new gun crossover position might be also different in X-Y plane, too. Spot size dependent gun shift folder is to compensate this change, e.g. realigning C1 and C2. The new alignment may be saved in FEG Register for the future use. When recalling a stored FEG setup, the gun and condenser alignments including the gun crossover position are resumed.

As a beam performing aperture, the alignment of the C2 aperture is critical since a misaligned C2 aperture results in a misaligned C2 to both C1 and C3. The reproducibility of the motorised

aperture on Titan is pretty high. Normally there is no need to mechanically align the C2 aperture when a new size is introduced. However, if it is necessary, it is recommended to align the C2 aperture in Free Control mode with C3 off.

The condenser system requires a high excitation as well as a low excitation of C3 lens to from either a small or a large parallel illumination in TEM mode. The hysteresis effect can produce a beam deflection during the area zoom. It is seen as slight beam movement and beam tilt. The simplest way to handle this is to set 'Normalize condenser' or 'Normalize all' to one of the user defined keys. When beam deflection is seen, for example during beam focusing, pressing the key will bring beam to its normalized position. Typically the beam shift reduces to an order of 10nm or better. Normalization can be also automatically executable after setting it in Normalization control panel. Then the condenser will be automatically normalized when it is idle for 0.5s after a change of the illumination.

Some simple tests can be performed to evaluate the goodness of the alignment of the condenser system. When changing spot size, the beam intensity increases or decreases while the illumination size should keep the similar. When changing magnification in TEM mode with Auto Zoom selected (see figure 7), the illumination size on view screen should be the similar (but not the real illuminated size on the specimen) and its position should not shift too much after condenser normalization. Besides, the displayed illuminated area size (see figure 6) should be close to the feature being illuminated, if the calibration in that particular mode has been done.

WORLD HEADQUARTERS
5350 NE DAWSON CREEK DRIVE
HILLSBORO, OREGON 97124 USA
PH: +1.503.726.7500

FEI EUROPE
PH: +31.40.23.56000

FEI JAPAN
PH: +81.3.3740.0970

FEI ASIA PACIFIC
PH: +86.21.6122.5988

fei.com

TUV Certification for design, manufacture, installation and support of focused ion and electron-beam microscopes for the NanoElectronics, NanoBiology, NanoResearch and Industry markets

FEI COMPANY™
TOOLS FOR NANOTECH

© 2008. We are constantly improving the performance of our products, so all specifications are subject to change without notice. The FEI logo and Tools for Nanotech are trademarks of FEI Company.

FEI COMPANY™
TOOLS FOR NANOTECH

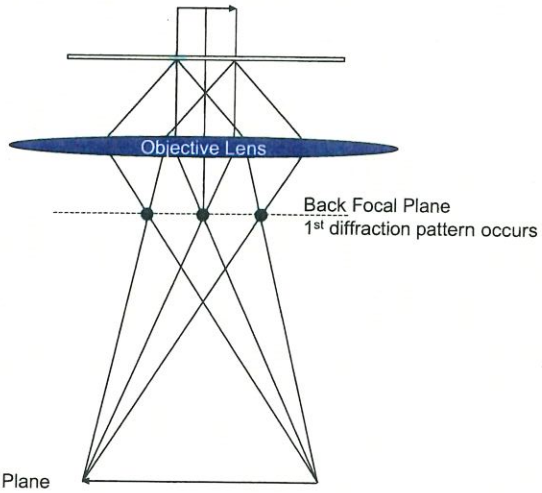
Module3
Electron Diffraction in TITAN

By Emrah Yucelen

1. Introduction to electron diffraction

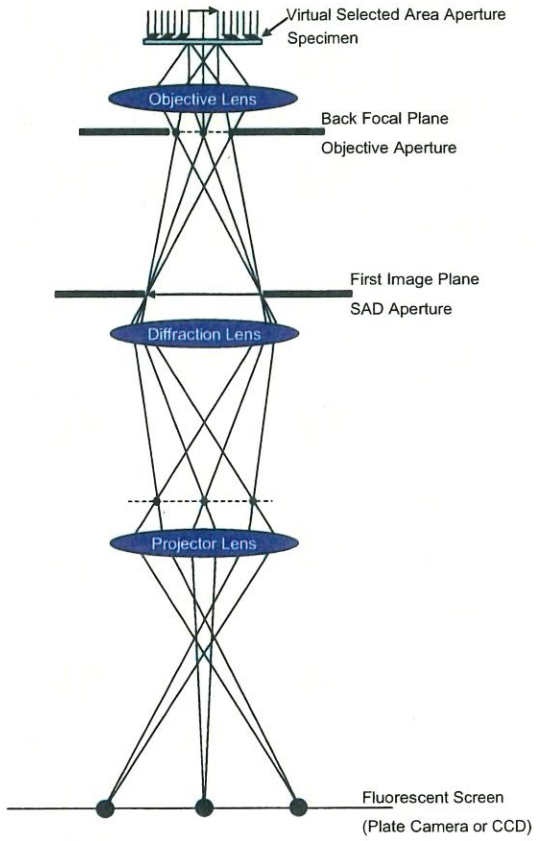
Owing to the negatively charged electrons and the short wavelength of the electron wave generated in a transmission electron microscope (e.g. from 80 kV to 300kV), the incident electron beam interacts strongly with atoms and is heavily scattered both elastically and inelastically. When the sample is crystalline, the incoming electron beam is collectively scattered or diffracted. In more precise terms, when the secondary waves that emerge from the points (atoms in this case) within the scattering volume (unit cell and crystal shape) interfere constructively along special directions, electron beam is said to be diffracted. The basic explanation of diffraction from crystalline materials is due to W. Bragg who showed and formulated that when path difference between the scattered waves from specific crystal planes is multiple number of $n\lambda$, the waves reinforce each other and strong intensity observed for a specific angle; $2d\sin\theta_B=n\lambda$ which is known as Bragg law. In this equation, n is an integer, d is the distance between reflecting (scattering) planes and θ_B is the angle to which strong intensity occurs. Detailed description and formulation of the electron diffraction can be found in the most TEM text books, such as those listed at the end of this module [1, 2, 3, 4].

Figure 1: Diffracted electron beams are going through the objective lens below the specimen. Diffracted beams from the same set of the atom planes are focused into a spot at the back focal plane. Electron Diffraction patterns are formed.



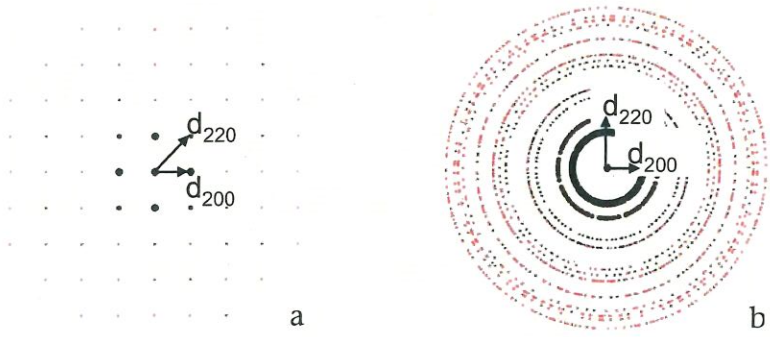
In a transmission electron microscope, diffracted electron waves are collected and focused by the objective lens which forms a diffraction pattern of the specimen at the back focal plane. Figure 1 schematically shows the function of the objective lens. When diffraction button on the control panel is active, the microscope is switched to diffraction mode. The ray path in diffraction mode is presented in Figure 2. In this mode, instead of the image plane of the objective lens, the imaging lens system (represented here as a diffraction lens plus a projector) is focusing at the back-focal plane of the objective lens.

Figure 2: Ray path in diffracted mode.



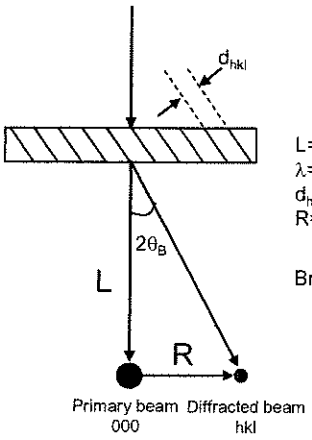
The diffraction pattern on the back-focal plane of the objective lens is further magnified by the imaging lens system. Depending on crystal structure, microstructure and texture of the sample, different electron diffraction patterns will be observed on the fluorescent screen. A regular pattern of spots from a single crystalline with FCC structure is demonstrated in Figure 3a. The spots represent lattice planes in the crystal and the distance to the pattern centre is the reciprocal of inter-planar distance.

Figure 3: Simulated electron diffraction patterns from a FCC crystal a) Spot pattern from a single grain or single crystals, b) ring pattern from poly-crystalline.



When the electron beam does not illuminate one single crystal but a number of randomly oriented micro-crystals, the diffraction pattern will consist of concentric circles, as illustrated in Figure 3b. In this case the radius R of the circles is related to a characteristic spacing d of lattice

plane in the crystal. Lattice spacings can be calculated if the camera length is known, as it is shown in Figure 4. The camera length is calibrated using samples of known lattice spacings, e.g. oriented gold crystal or minerals. Information on how to determine d-spacings and how to extract more information from spot/ring diffraction patterns are greatly covered in references [1,2,3,4].



L=camera length
λ= wavelength of electron wave
d_{hkl}=interplanar distance
R=distance measured between primary and diffracted spots

Bragg Law: $\lambda=2d\sin\theta_B$

$R=L\tan 2\theta_B$

$Rd_{hkl}=L\lambda$

Figure 4: Demonstration of Bragg law in transmission geometry.

Normally used electron diffraction techniques in materials research are

- Selected area electron diffraction (SAED): to obtain spot electron diffraction patterns using (near) parallel electron beam.
- Convergent beam electron diffraction (CBED): to obtain disk electron diffraction pattern using focused electron beam.
- Nano beam electron diffraction (NBED): to obtain spot(-like) diffraction patterns from nanometre area without using the selected area aperture.

In the following sections details of those techniques will be discussed.

2. Diffraction with/without a selected area aperture

In transmission electron microscopes with the conventional two condenser lens system, such as Tecnai series, parallel illumination is only achieved at a particular condenser focus or a specific, C2 aperture dependent, size of the illuminated area. Typically this size is more than a few microns. Away from this particular condenser focus, the electron beam starts gradually to either converge or diverge. In both the cases, the incident electron beam is not parallel, away form the basic request for forming a spot diffraction pattern. The size of spots grows with the increase of the beam inclining angle, so that spots become discs and, further, overlapped discs.

Since a spot pattern from an area smaller than the specific size mentioned above could not be directly obtained, a selected area aperture, which is located at the 1st image plane (Figure 2), was introduced. We can keep a parallel illumination (or, with certain tolerance, near parallel) that illuminates a relatively large area. A selected area aperture is inserted, selecting part of the specimen from its 1st image, to form a spot diffraction pattern. Because of the parallelism of the incident beam, the diffraction pattern is a spot pattern (Figure 5). Since the objective lens magnification is typically 30 times, a 10μm SA aperture will set a specimen area of 0.33μm to form a diffraction pattern. In other words, if an aperture is inserted in the objective image plane, it is equivalent to inserting an aperture in the plane of the specimen, but smaller by a factor of the objective lens magnification.

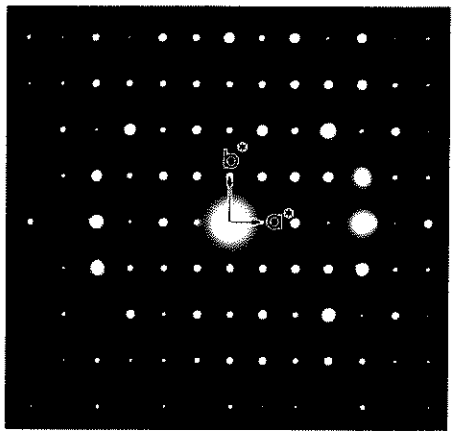


Figure 5: Selected area diffraction pattern of an intermetallic compound in an Al alloy, obtained using a selected are aperture. Arrows indicate vectors in reciprocal 2D space.

Instead of using a SA aperture to define an area to form diffraction patterns, illumination itself may also be an area selector. As we mentioned above, it may introduce the beam inclining angle and change a spot pattern to a disc pattern. The effect is much weaker in Titan because its 3-condenser system may keep a parallel illumination within a large range, typically from a few tens of microns to a few hundreds of nanometres. Besides, without using a SA aperture has several advantages over using an aperture. Firstly, diffraction patterns can be obtained from a smaller area than the SA aperture can provide. Secondly, the optical conditions are closer to those in imaging, especially for high-resolution work. Thirdly, the so-called diffraction error due to spherical aberration of the objective lens, which makes the area outside the SA aperture contribute to the diffraction pattern, does not exist.

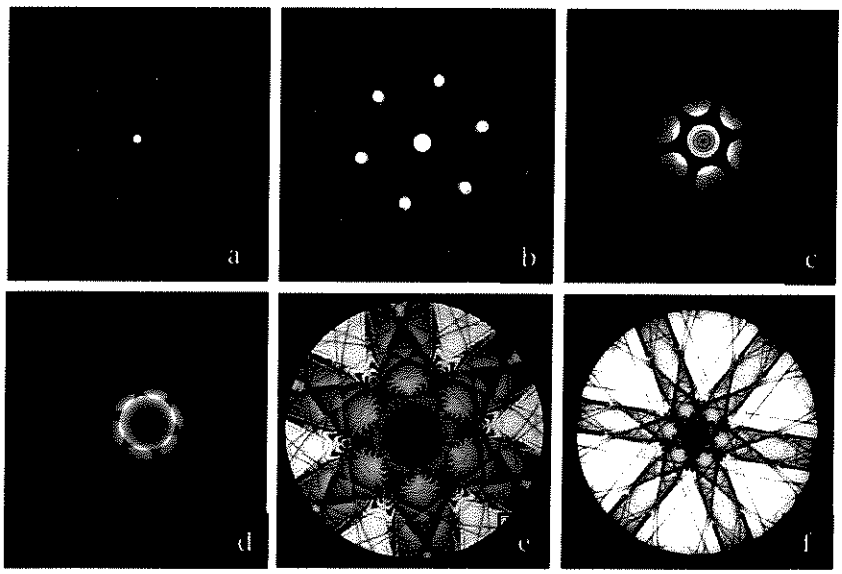


Figure 6: Electron diffraction patterns of Si <111> with different beam convergence angles. The convergence angle increases from a) to f). a) an SA pattern or a spot pattern; b) pattern with small convergence angle (termed spot-like pattern for convenience); c) and d) convergent beam electron diffraction patterns; e) and f) large angle convergent beam electron diffraction patterns.

The side effect of using beam as the area selector, especially for a conventional two condenser system, is the increase of the convergence angle as described above. The evolution of diffraction pattern with the convergence angle is illustrated in Figure 6. If the convergence angle is reasonably small, the spots become small discs but without intensity variations within them (Figure 6b). For simplicity, we call it a spot-like pattern because it is similar, although not exactly, to a real spot pattern in Figure 6a. When the convergence angle is large, the diffraction becomes

convergent beam electron diffraction (CBED). In CBED, the intensity variations within the discs, that contain rich crystalline information of specimen, are seen. In Figure 6c, the convergence angle is small enough so that the discs are not touching each other; while in Figure 6d the discs start to overlap. Now one needs to separate one disc from the others, if further increase of convergence angle is needed, in order to see the clear intensity variation within a particular disc. The technique is called large angle convergent beam electron diffraction (LACBED) as shown in Figure 6e and 6f.

3. Diffraction with parallel illumination on Titan

Titan three-condenser lens design of Titan offers a great flexibility. One of its major advantages is to obtain spot diffraction patterns (or ring patterns if polycrystalline) without using the selected area aperture. In principle, one does not need a SA aperture and the SA aperture lost its basic function as in the conventional TEMs. Once the condenser zoom is properly aligned and calibrated, the range of the parallel illumination is displayed in TEM mode that varies with the C2 aperture size (Figure 7). To obtain a spot/ring pattern is just straight forward by adjusting the beam to illuminate only the region of interest in TEM mode and then switching to diffraction mode (Figure 8). The operator should only pay attention to if the size of interest is outside the zooming range as indicated in Figure 7. If yes, changing to a different C2 aperture size probably helps.

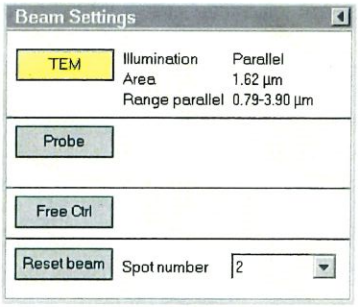
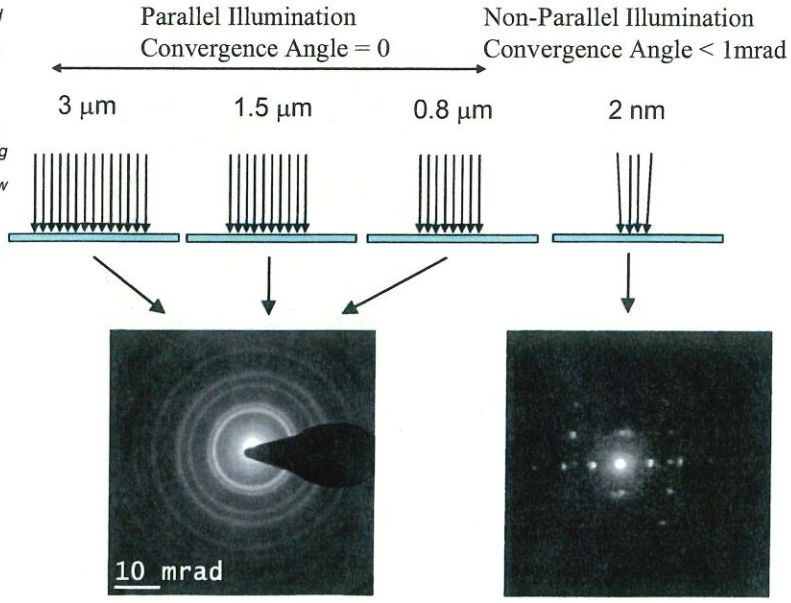


Figure 7: In the Beam Settings control panel, the range of parallel illumination is displayed.

Figure 8: Diffraction patterns from a gold cross-grating in TEM mode. The ring pattern occurs within a large range of illuminated area. When the beam is slightly convergent, the pseudo spot-like pattern (non-overlapping discs) is obtained from an area down to a few nanometers.



Apart from the area size, the C2 aperture size also defines the maximum beam convergence angle. Both the maximum beam convergence angle and the minimal parallel illuminated area size increases with the C2 aperture size. The smallest area of the parallel illumination can be achieved with the smallest C2 apertures, which is typically 50μm for a standard Titan configuration. In theory, this smallest parallel illuminated area is about 180nm with the 50μm C2 aperture. The optional smaller C2 aperture (10 or 20μm) can be ordered upon request, so that the smallest area could be reduced further to only a few tens of nanometres. On the other hand, if we accept slight deviating from the perfect parallelism, it is possible to obtain spot-like patterns from an area of only a few nanometres (Figure 8).

Nowadays, people are paying more attention to the diffraction from small area down to nanometre range, such as a single nano-particle or a single carbon nanotube. This nano beam diffraction technique (as well as its predecessor of μ-diffraction, referring to diffraction from a micron size area) requests not only a small illuminated area but also the parallelism of the beam. Thanks to its three-condenser system, to do NBED is not difficult on a Titan. To achieve a nanometre size parallel probe, it is recommended to use a small C2 aperture as well as the strongly demagnified probe, i.e. high gun lens (6-8) and large spot size number (6-9), in microprobe.

The steps to obtain a diffraction pattern with parallel (or nearly parallel) illumination are as the followings:

1- In TEM mode, find the area of interest and place the region of interest (particle, precipitates, interface etc) on the centre of the small fluorescent screen (if for NBED at a reasonably high magnification) (Figure 9a)
2- If necessary and available load a stored FEG Register, i.e. high-gun lens and high spot number (it is recommended to have a re-aligned gun for optimal performance)
3- Insert and align a C2 aperture, making sure the size of the interest feature is within the area zoom range (Either the largest or the smallest C2 aperture needs to be used if the feature is over/under the area zoom range)
4- Focus beam using Intensity knob until the beam reaches the size of the interest feature (Figure 9b)
5- Use beam shift to put focused beam on the centre of the small fluorescent screen. Stigmate the beam using Condenser Stigmator (accessible from Stigmators OCX)
6- Switch to diffraction mode and observe diffraction pattern on the fluorescent screen
7- Adjust camera length according to the recording media
8- If it is necessary stigmate diffraction pattern using Diffraction Stigmator (accessible from Stigmators OCX)

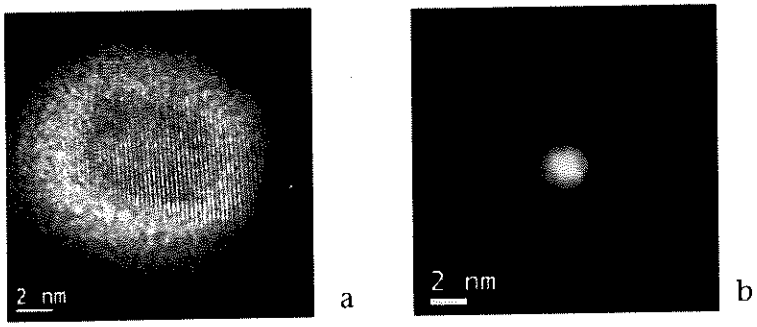


Figure 9: A gold nano-particle (a) and a focus ~2nm probe for NBED (b).

Please note in certain range, typically a few hundred nanometres, the real parallel illumination is formed by nanoprobe TEM. Please refer to Condenser System module for details. When doing NBED, the beam current is too low for successful imaging. Therefore as an alternative way, microprobe STEM mode can be used for acquiring both image and diffraction patterns. This

special mode is different from the standard STEM in the convergence angle.

It is also worth mentioning that even a disc-like diffraction patterns can be focused to something like a spot pattern by refocusing the diffraction lens (FOCUS knob in diffraction mode). This is not a real spot pattern due to the parallel incoming electron beam, and the pattern is not formed on the back focal plane of the objective lens. To verify this, an objective aperture can be used in diffraction mode. Focus the diffraction lens to the back focal plane means the edge of the objective aperture is sharp. If beam is parallel, the diffraction is sharp after accurately focusing the diffraction lens (Figure 10).

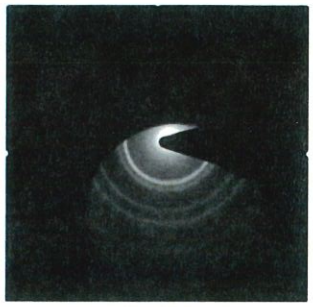
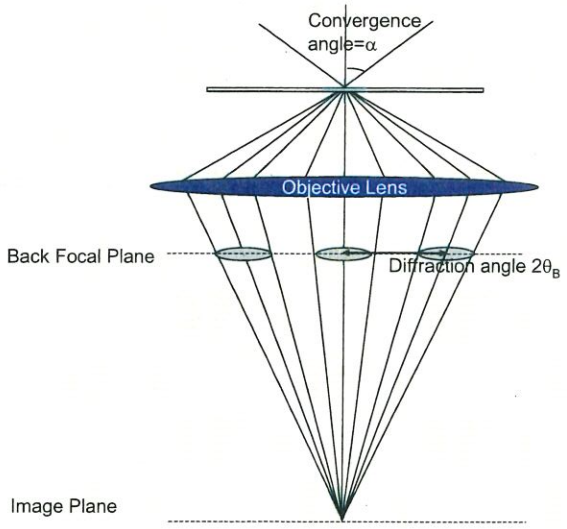


Figure 10: Objective aperture is inserted and focused at the back focal plane.

4. Convergent beam electron diffraction

Figure 11: Formation of CBED pattern. Each beam within the illumination cone acts as an incident beam. Therefore, discs are formed on the diffraction plane.



When the incident beam is converged, CBED pattern forms as outlined in figure 11. When the convergence angle is very low, the diffraction disc is so small that it is actually a spot-like pattern. When the convergence angle is higher, the discs also become larger in size and the intensity variation appears within the discs. If the convergence angle is even higher, the discs become so large that they overlap each other. Therefore, the appearance of a CBED pattern depends on the beam convergence angle, which is in turn determined by the condenser system setting and the C2 aperture size.

In Titan CBED patterns can be obtained in both nanoprobe Probe mode and microprobe Probe.

By default, probe mode is nanoprobe setting where the convergence angle ranges from the medium to the high. When 50μm C2 aperture is used the lowest convergence angle in Probe mode is about 6mrad when the beam is fully focused. As a result of convergence zoom, the convergence angle can be increased until about 16mrad (Figure 12a). In most cases, this convergence angle value is already too high to keep the diffraction discs separated, even with a 50μm C2 aperture. For example, in silicon <110> zone axis, the closest spots (111) are 6.4mrad away from the central spot at 300kV. The (111) discs are just separated from the central disc at the lowest convergence angle. When the convergence angle increases to 11mrad, (111), (200) and (220) discs will overlap with the central disc, as what you can see in Figure 12b that is a typical image in STEM mode.

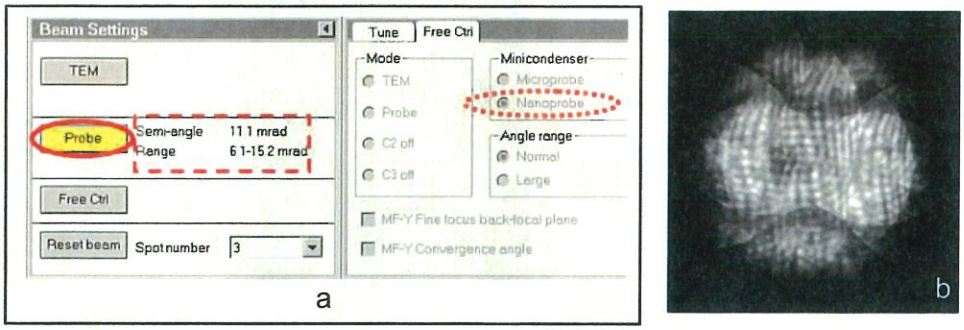


Figure 12: The convergence angle range is from 6mrad to 15mrad in nanoprobe Probe mode with a 50μm C2 aperture (a). Generally, the convergence angle is too high to keep the individual discs separated. (b) is obtained in a standard STEM setting.

The convergence angles in Probe mode can be reduced if the smaller C2 apertures are available. For example, a 20μm C2 aperture will create a minimal convergence angle of about 2.5mrad (Figure 13a). If the aperture applies to a sample of silicon <111>, all the discs will be well separated (Figure 13b).

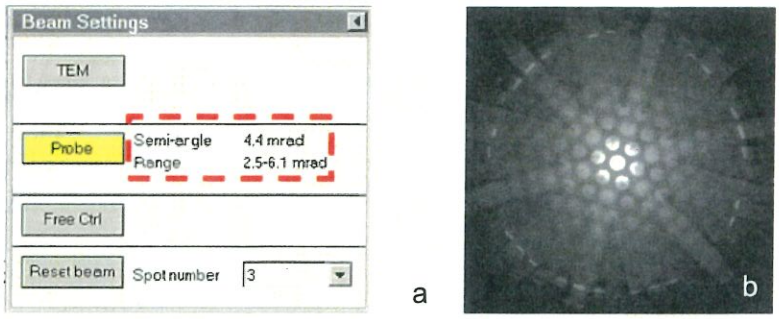


Figure 13: The convergence angle range becomes from 2.5mrad to 6mrad in nanoprobe Probe mode if a 20μm C2 aperture is used (a). The same diffraction discs as in Figure 12b will be well separated (b).

Alternatively, microprobe Probe mode, a special setting on TITAN, can be used for CBED with relatively small convergence angles. In this setting, microscope is in Probe mode with minicondenser lens is optically switched on. The beam convergence angle in microprobe Probe setting is about 5 times less than that in nanoprobe Probe mode. To enter microprobe Probe mode, one needs go to Free Ctrl in Beam Settings and select both Probe and Microprobe (Figure 14). The convergence angle range with a 50μm C2 aperture is from 1.3 to 4.8mrad, compared 6.1-15.2mrad in nanoprobe Probe mode with the same size of the C2 aperture. The disadvantage of the microprobe Probe mode is that the spot size is about 5 times larger than that

in the nanoprobe Probe mode.

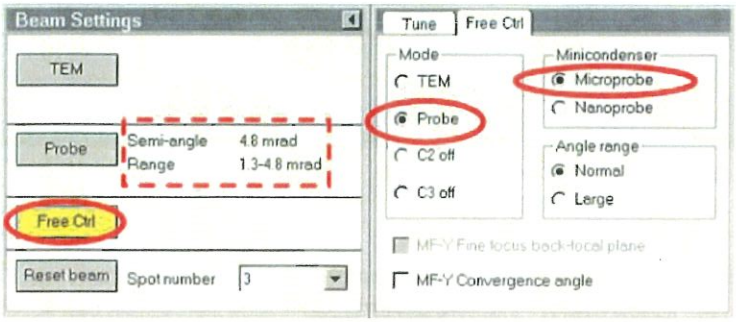


Figure 14: To set microprobe Probe mode in Beam Settings OCX of Titan. A 50μm C2 aperture produces the lowest convergence angle of 1.3mrad.

On the other hand, to obtain the highest convergence angle for LADBED, we need the largest C2 aperture and working in nanoprobe Probe mode. When MF-Y Convergence angle in Free Ctrl (see Figure 14) is selected, the multifunction-Y knob can be used to fine tuning the convergence angle. Besides, if Large radio button in Angle Range frame is selected (see Figure 14), the convergence angle becomes even larger, well suitable for LACBED.

5. How to determine convergence angle of a diffraction patterns

We have encountered many time in this module the convergence angle, either it is small for spot/ring diffraction patterns or it is large for LACBED patterns. Experimentally, the convergence angle may be determined by measuring the diffraction disc size comparing to the known d-sapcing on the same diffraction pattern. Of course, if the convergence angle is too small the measurement will be less reliable due to the spreading effect of the recording media, the shape effect of the finite crystal size etc. The best situation is when the large diffraction discs are separated and clearly measurable, such as the example of a CBED pattern of silicon <111> in Figure 15.

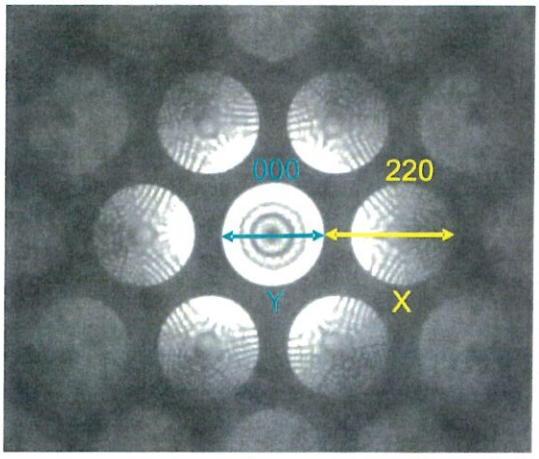


Figure 15: CBED pattern of Si <111> that can be used for the measurement of the convergence angle.

In Figure 15, the central spot is surrounded by six {220} diffraction discs. 2 times of Bragg angle of 220 planes at 300kV is 10.26mrad. It can be immediately seen that the convergence angle is

smaller than 10.26/2=5.13mrad since the diameter of the discs is smaller than the diffraction angle ($\alpha<\theta_B$). With the measurements of X and Y in Figure 15, to calculate the convergence angle is simply the ratio of Y/X multiplied by the Bragg angle. The measured convergence angle in the example is about 4.08mrad.

References

1. Transmission Electron Microscopy; Text Book for Materials Science, D. B. Williams, C. B. Carter, Plenum Press, 1996.
2. Transmission Electron Microscopy, L. Reimer, Springer-Verlag, 1997.
3. Introduction to Conventional Transmission Electron Microscopy, M. De Graef, Cambridge University Press, 2003.
4. Transmission Electron Microscopy and Diffractometry of Materials, B. Fultz, H. Howe, Springer-Verlag, 2003.

FEI COMPANY™
TOOLS FOR NANOTECH

WORLD HEADQUARTERS
5350 NE DAWSON CREEK DRIVE
HILLSBORO, OREGON 97124 USA
PH: +1.503.726.7500

FEI EUROPE
PH: +31.40.23.56000

FEI JAPAN
PH: +81.3.3740.0970

FEI ASIA PACIFIC
PH: +86.21.6122.5988

fei.com

TUV Certification for design, manufacture, installation and support of focused ion- and electron-beam microscopes for the NanoElectronics, NanoBiology, NanoResearch and Industry markets

© 2008. We are constantly improving the performance of our products, so all specifications are subject to change without notice. The FEI logo and Tools for Nanotech are trademarks of FEI Company.

FEI COMPANY™
TOOLS FOR NANOTECH

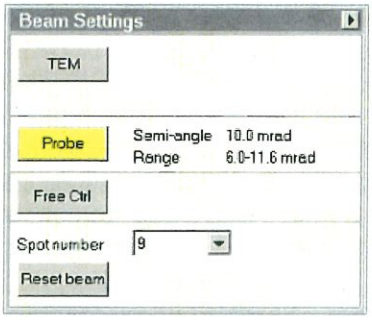
Module 4
High-Resolution STEM

By Michel van der Stam and Joerg Jinschek

In Titan, the optical system has been fully optimized to allow the microscope to reach the limits of its optical resolution. Current noise in the lenses and deflectors has been brought to an absolute minimum, so that it its influence on the resolution is minimized. Achievable resolution depends therefore mostly on a proper alignment of the entire system. The purpose of this module is to describe the settings and alignments procedures in STEM, thereby focusing towards reaching the properly aligned state for high-resolution.

1. The basic optical scheme for HR-STEM

HR-STEM on Titan is performed with the optical system in Probe mode - a mode which closely resembles Tecnai's HR-STEM mode.



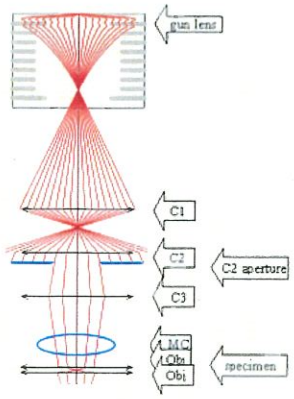


Figure 1: Overview of Titan's condenser setting in "Probe mode" (minicondenser lens "MC" is off)

Figure 1 shows the positions and settings of the lenses of the condenser system in Probe mode. In order to obtain a probe with low lens aberrations, "normal range" option (referring to the range of convergence angles) is the preferred mode. In this setting there is no beam cross-over between the C2 and the C3 lens, which is optimized for the optical quality of the focused probe. A weakly exited C2 lens in the "normal range" option contributes only minimal to the total probe performing spherical aberration.

In Probe mode the minicondenser lens is optically off (so-called nanoprobe mode) since this gives maximum demagnification of the probe on the sample. Thereby smallest spots (electron probe) can be achieved, as well as the impact of disturbances coming from the condenser system will be minimized.

2. Microscope settings

For HR-STEM it is very important to use a bright electron source. Therefore, at 300kV the extraction voltage of the FEG should be set to 4.5 kV. A typical gun lens setting is 6 (see Figure

2). The increased energy spread from the gun with these settings does not have a significant impact on the STEM resolution. After switching the FEG to its correct settings, one has to execute the gun alignment procedures. However, if a saved FEG setting is resumed from FEG register, the gun re-alignment is generally not really necessary.

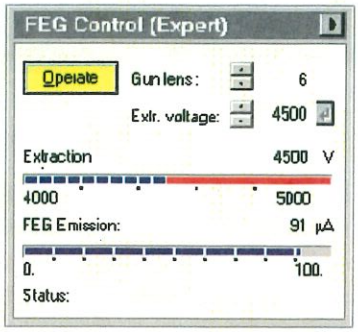


Figure 2: FEG control panel for the typical HR-STEM setting.

It is of critical importance that the Titan 3-lens condenser system is well aligned and is not changed after the condenser system has been aligned. The condenser alignment depends on the gun settings (high tension, extraction voltage, gun lens etc). Changing one of these parameters would result in the different positions (XY and height) of the gun cross-over. Fortunately, once it is aligned and saved in the FEG register, it can be reused quickly. You may find more information about the condenser system alignment in the relevant module.

It is also very important to keep in mind that the condenser alignment relies on the condenser 2 aperture position. The 50 µm C2 aperture needs to be centred on the axis between the C2 and the C3 lens in Probe mode. After executing the "Direct Alignments" procedures "Beam shift" and "Rotation centre", the position of the C2 aperture needs to be optimized using the "Aperture" control window. The C2 aperture is centred when the spot condenses and expands concentrically while defocusing the probe using the Intensity knob.

As critical as the condenser system alignment is the specimen height (z-height setting of the compustage) for optimal operation of the optics. It influences on how much of the objective lens' magnetic field is below the specimen and/or above the specimen. The optimum specimen height is at the eucentric plan, or the middle plane within the objective lens gap. If this is not true, e.g. the specimen is closer to the lower pole piece the current in the upper part of the objective lens becomes higher in order to keep the image in focus. At the same time, the condenser system would become weaker, or focus the probe less. This changes the optics ray paths between the condenser system and the objective lens. The main effect will be away from the fine tuned optimum probe convergence angle.

After the compustage z-height has been set correctly, the probe convergence-angle can be varied around its nominal value within a limited range by using the flexibility of the condenser system. Proper fine-tuning of the probe convergence-angle is essential to get to the HR-STEM resolution. It can be achieved through MF-Y knob after selecting "MF-Y Convergence angle" in Beam Settings control panel (Figure 3).

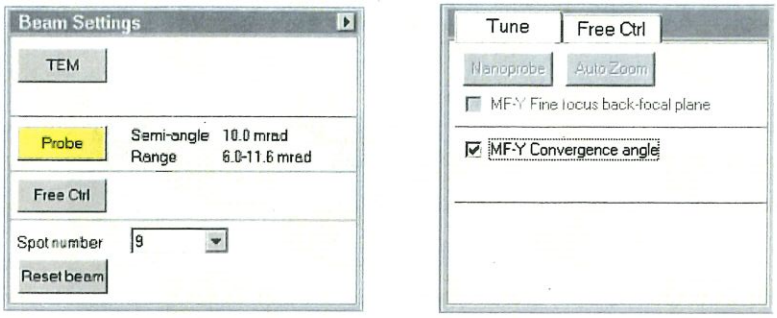


Figure 3: The Beam Settings control panel in Probe mode and its flap-out page of Tune.

Three parameters will influence the size of the electron probe, and therefore the optimum setting of the probe convergence angle for HR-STEM.

- The probe's convergence-angle in association with the spherical aberration C_S of the objective lens causes a blur of the probe. The sizes of the blur disc (D_{Cs}) increase with the convergence angle.
- The size of the C2 aperture, which also determines the convergence angle, has another effect. This so-called diffraction effect creates a disk with a size of blur D_{diffr}. The disk size is proportional to the wavelength and inversely proportional to the convergence angle.
- The more the condenser system demagnifies the FEG tip onto the sample the smaller the image of this FEG tip becomes (size of FEG tip image = D_{geo}).

Then the total probe diameter D_{tot} is simply the combination of all of these contributions:

$$D_{tot} = D_{geo} + D_{Cs} + D_{diffr}$$

All three contributions need to be balanced to obtain the optimum probe setting.

- The stronger the gun lens and the condenser system demagnify the source, the smaller the size of the image of the FEG tip will be on the sample. This image D_{geo} can therefore be decreased by increasing gun lens number and/or spot number. The side effect of this demagnification however is that more and more current is cut away from the electron probe. Consequently, the best probe with the highest resolution will be at very low beam current. As low beam current is not very practical due to poor signal-to-noise ratio, there needs to be a balance so that the geometrical probe size D_{geo} is small enough to get resolution and is large enough to have sufficient current in the probe.
- Even if the image of the FEG source D_{geo} would be made infinitely small, this does not result in a sub-angstrom spot because the spherical aberration disc D_{Cs} and the diffraction disc D_{diffr} can not be made small simultaneously, considering their relationship with the convergence angle.
- D_{Cs} increases rapidly with increasing probe convergence angle. In contrast, the diffraction blur D_{diffr} gets smaller with increasing probe convergence angle. By the way, the chromatic aberration does not play a significant role here! It turns out that by selecting a decent small source image D_{geo} (with gunlens number = 6 and spot number = 9) and summing up all other contributions the smallest total probe diameter D_{tot} will be reached when setting the optimum probe convergence angle for Titan 300kV SuperTwin (without probe C_S corrector) to approximately 10 mrad when using the C2 aperture of 50 µm.

The strategy to set the probe convergence angle to the correct value is to first set it to 10 mrad directly using the flap-out Tune page of Titan's Beam Settings panel (see Figures 3), and then to use the Ronchigram for double checking and fine tuning of the probe convergence angle.

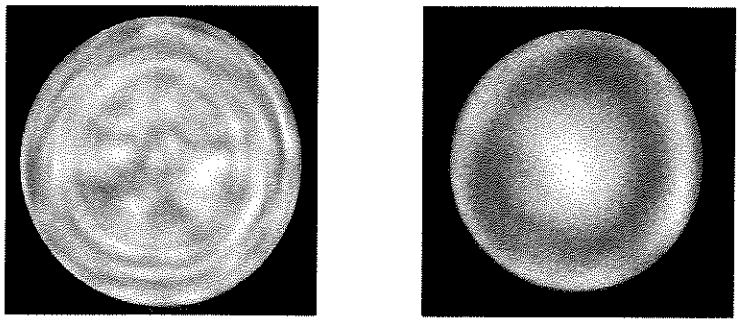


Figure 4: Ronchigrams (a) not at the best probe focus and (b) at the best focus with a right convergence angle.

A Ronchigram (Figure 4) is the shadow image of the probe illuminating the specimen, in diffraction mode. The procedure to observe a Ronchigram is:

- set specimen at eucentric height and focus image at eucentric plane (Eucentric Focus)
- select an amorphous area
- focus the probe to the sample with Intensity
- switch to diffraction
- fine tune probe focus (Intensity knob) until there are no details of the sample in the central part of the Ronchigram (termed "blow-up point")
- in case the Ronchigram is not circular symmetric, astigmatism needs to be corrected by using "Condenser stigmator"

The circular intensity variations in Figure 4a and Figure 4b are due to the spherical aberration. The dark rings at the centre of the Ronchigram now define visually how to tune the probe convergence angle for optimum HR-STEM resolution. The edge of the 50 µm C2 aperture should be at the bright ring just outside the first black ring around the central area, as is shown in Figure 4b. In case the beam limiting 50 µm C2 aperture does not cut-off the Ronchigram at the right position with a small difference, then the probe convergence angle needs to be optimize by changing the convergence angle in Tune page (Figure 3).

For reaching HR-STEM resolution it is required to have zero condenser two-fold astigmatism, and low three-fold condenser astigmatism. Both can be done by selecting the proper option in the Stigmator panel. Because of tolerances in the condenser system, the electron beam might pass off-axis through the Condenser C2 stigmator, resulting in a slight beam tilt inside the stigmator when doing stigmation. Therefore the "Rotation centre" of the objective-lens/condenser needs to be checked in Direct Alignments.

Here it is worth mentioning that beam tilt and beam shift need to be set correctly in Probe mode, too. To check the beam tilt, set the "Beam tilt pivot points" in Direct Alignments correctly with the specimen at the preset eucentric focus position in Probe mode. After confirming the correct C2 aperture position (as described above), the rotation centre needs to be carefully adjusted (probe concentrically in and out). This is the rotation centre of the upper + lower pole piece of the

objective lens. It will be discussed later, how to optimize the rotation centre further so that it is optimized on the upper pole piece of the objective lens only for an optimized probe.

The "Align Beam Shift" in "Direct Alignments" gives the possibility to make the beam jump to the centre of the screen after resetting the user beam shift. This needs to be done at a relatively high SA magnification.

3. (HR-)STEM mode

The specimen needs to be at the correct (eucentric) height. This is the imaging plane after pressing the "Eucentric focus" button. It needs to be checked if the objective lens current does not change when switching from Probe mode to STEM by looking at the objective lens (Obj %) reading on the Titan software display bar.

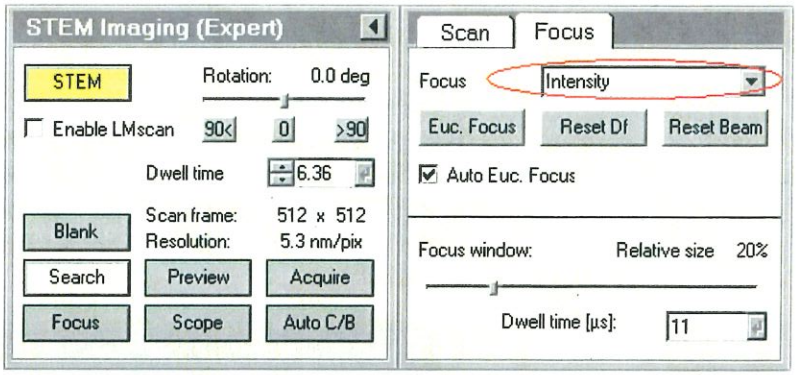


Figure 5: STEM Imaging control panel.

On a standard (non probe corrector) system, in STEM the focus knob is linked to the C3 lens, if the Focus selection is Intensity (figure 5). "Auto Euc. Focus" needs to be selected to make sure that the objective lens focus current in HM-STEM is at its preset eucentric value when switching to STEM. It is also very important to "Normalize all" the lenses, which can be set to a user defined button when switching the mode, and to press "Reset Beam" key in the Focus flap-out page (Figure 5) to check the beam shift.

For optimum HR-STEM imaging the rotation centre of the objective lens and the C2 aperture position need to be checked, and maybe be fine-tuned. The position of the C2 aperture determines, which part of the focusing condenser lens is *in use*. Ideally, the C2 aperture position is tuned such that the central (on-axis) part of the focusing condenser lens is used. The correct rotation centre setting makes sure that the beam (that exits on-axis the condenser lens) enters on-axis the objective lens. When the C2 aperture position is known to be correct, fine tuning the rotation centre can be done by using the "Direct Alignments". If the image of the C2 aperture in the Ronchigram is not centred on the symmetry centre then there is a mismatch between the aperture position and the rotation centre. After re-checking the position of the C2 aperture as has been described previously, the "Rotation centre" should be fine-tuned using the "Direct Alignments" procedure.

For optimal image contrast it is important to choose the right camera length and the correct HAADF detector settings. Using a standard Titan 300kV SuperTwin system, camera length of around 200mm is optimal for HAADF contrast (Z-contrast). Higher camera lengths would include Angular Dark Field contrast (the reverse of bright field contrast) into the STEM image. Brightness

level of the HAADF detector should be set to about 43%, and the Contrast level typically to values between 30% and 70% (Figure 6).

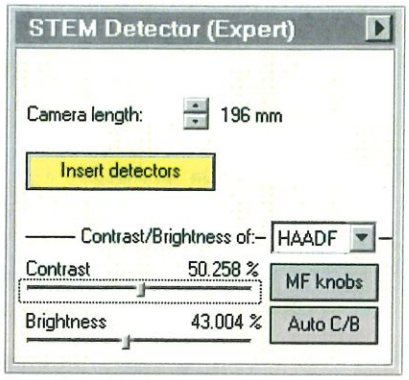
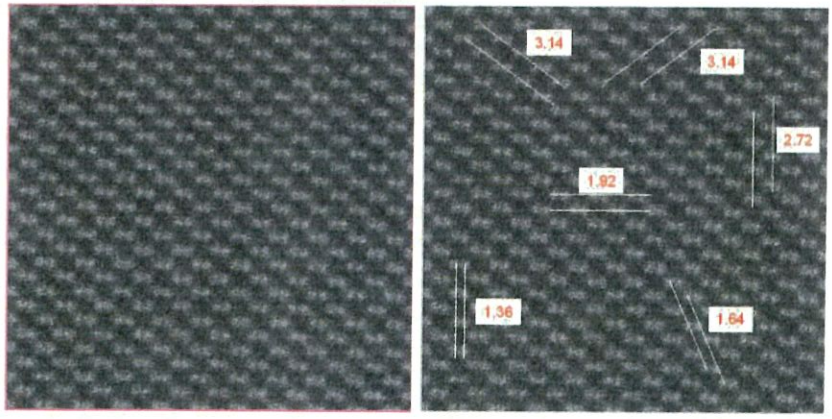


Figure 6: STEM Detector control panel.

4. HR-STEM imaging

The specimen typically used for HR-STEM proof-of-performance is the silicon specimen perfectly aligned in <110> zone axis. It shows (line) spacing of 3.1Å, 2.7Å, 1.9Å, 1.6 Å, and 1.36Å (dumbbell distance). These spacing are quite helpful in terms of fine-tuning focus and astigmatism, as well as specimen position and specimen tilt even before obtaining the final resolution.

Figure 7: HR-STEM image of silicon <110> specimen



An area on the silicon sample which has the right quality (right thickness, thin/no amorphous layer, correct zone axis orientation) needs to be selected. Leave diffraction mode, normalize all lenses and focus the probe (intensity). Switch back to diffraction mode and move the sample in focus (correct z-height) with compustage-z ("blow up" point in Ronchigram).

The intensity ratio between bright field disc and scattered discs is a helpful criterion to determine if the specimen has the correct thickness. They should have approximately the same intensity. If the sample is too thin than the intensity of the bright field discs is much stronger than the scattered discs, therefore there will be no sufficient signal on the detector to resolve 1.36Å. If the sample is too thick the intensity of the bright field disc will be weak. Then electrons will scatter multiple times in the sample thus destroying resolution.

FEI COMPANY™
TOOLS FOR NANOTECH

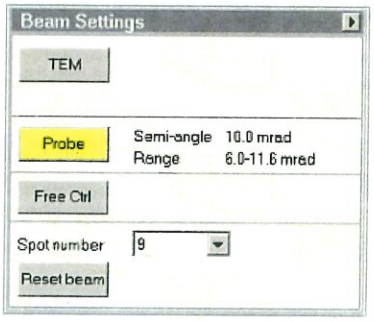
Module 4
High-Resolution STEM

By Michel van der Stam and Joerg Jinschek

In Titan, the optical system has been fully optimized to allow the microscope to reach the limits of its optical resolution. Current noise in the lenses and deflectors has been brought to an absolute minimum, so that it its influence on the resolution is minimized. Achievable resolution depends therefore mostly on a proper alignment of the entire system. The purpose of this module is to describe the settings and alignments procedures in STEM, thereby focusing towards reaching the properly aligned state for high-resolution.

1. The basic optical scheme for HR-STEM

HR-STEM on Titan is performed with the optical system in Probe mode - a mode which closely resembles Tecnai's HR-STEM mode.



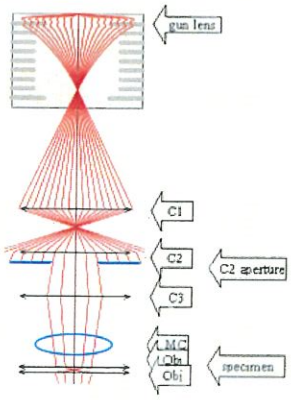


Figure 1: Overview of Titan's condenser setting in "Probe mode" (minicondenser lens "MC" is off)

Figure 1 shows the positions and settings of the lenses of the condenser system in Probe mode. In order to obtain a probe with low lens aberrations, "normal range" option (referring to the range of convergence angles) is the preferred mode. In this setting there is no beam cross-over between the C2 and the C3 lens, which is optimized for the optical quality of the focused probe. A weakly exited C2 lens in the "normal range" option contributes only minimal to the total probe performing spherical aberration.

In Probe mode the minicondenser lens is optically off (so-called nanoprobe mode) since this gives maximum demagnification of the probe on the sample. Thereby smallest spots (electron probe) can be achieved, as well as the impact of disturbances coming from the condenser system will be minimized.

2. Microscope settings

For HR-STEM it is very important to use a bright electron source. Therefore, at 300kV the extraction voltage of the FEG should be set to 4.5 kV. A typical gun lens setting is 6 (see Figure

2). The increased energy spread from the gun with these settings does not have a significant impact on the STEM resolution. After switching the FEG to its correct settings, one has to execute the gun alignment procedures. However, if a saved FEG setting is resumed from FEG register, the gun re-alignment is generally not really necessary.



Figure 2: FEG control panel for the typical HR-STEM setting.

It is of critical importance that the Titan 3-lens condenser system is well aligned and is not changed after the condenser system has been aligned. The condenser alignment depends on the gun settings (high tension, extraction voltage, gun lens etc). Changing one of these parameters would result in the different positions (XY and height) of the gun cross-over. Fortunately, once it is aligned and saved in the FEG register, it can be reused quickly. You may find more information about the condenser system alignment in the relevant module.

It is also very important to keep in mind that the condenser alignment relies on the condenser 2 aperture position. The 50 µm C2 aperture needs to be centred on the axis between the C2 and the C3 lens in Probe mode. After executing the "Direct Alignments" procedures "Beam shift" and "Rotation centre", the position of the C2 aperture needs to be optimized using the "Aperture" control window. The C2 aperture is centred when the spot condenses and expands concentrically while defocusing the probe using the Intensity knob.

As critical as the condenser system alignment is the specimen height (z-height setting of the compustage) for optimal operation of the optics. It influences on how much of the objective lens' magnetic field is below the specimen and/or above the specimen. The optimum specimen height is at the eucentric plan, or the middle plane within the objective lens gap. If this is not true, e.g. the specimen is closer to the lower pole piece the current in the upper part of the objective lens becomes higher in order to keep the image in focus. At the same time, the condenser system would become weaker, or focus the probe less. This changes the optics ray paths between the condenser system and the objective lens. The main effect will be away from the fine tuned optimum probe convergence angle.

After the compustage z-height has been set correctly, the probe convergence-angle can be varied around its nominal value within a limited range by using the flexibility of the condenser system. Proper fine-tuning of the probe convergence-angle is essential to get to the HR-STEM resolution. It can be achieved through MF-Y knob after selecting "MF-Y Convergence angle" in Beam Settings control panel (Figure 3).

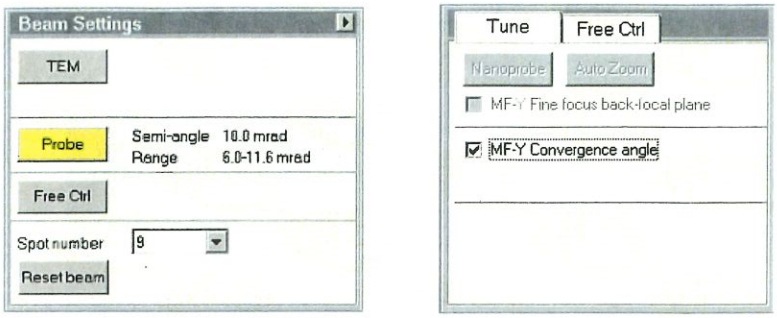


Figure 3: The Beam Settings control panel in Probe mode and its flap-out page of Tune.

Three parameters will influence the size of the electron probe, and therefore the optimum setting of the probe convergence angle for HR-STEM.

- The probe's convergence-angle in association with the spherical aberration C_S of the objective lens causes a blur of the probe. The sizes of the blur disc (D_{Cs}) increase with the convergence angle.
- The size of the C2 aperture, which also determines the convergence angle, has another effect. This so-called diffraction effect creates a disk with a size of blur D_{diffr}. The disk size is proportional to the wavelength and inversely proportional to the convergence angle.
- The more the condenser system demagnifies the FEG tip onto the sample the smaller the image of this FEG tip becomes (size of FEG tip image = D_{geo}).

Then the total probe diameter D_{tot} is simply the combination of all of these contributions:

$$D_{tot} = D_{geo} + D_{Cs} + D_{diffr}$$

All three contributions need to be balanced to obtain the optimum probe setting.

- The stronger the gun lens and the condenser system demagnify the source, the smaller the size of the image of the FEG tip will be on the sample. This image D_{geo} can therefore be decreased by increasing gun lens number and/or spot number. The side effect of this demagnification however is that more and more current is cut away from the electron probe. Consequently, the best probe with the highest resolution will be at very low beam current. As low beam current is not very practical due to poor signal-to-noise ratio, there needs to be a balance so that the geometrical probe size D_{geo} is small enough to get resolution and is large enough to have sufficient current in the probe.
- Even if the image of the FEG source D_{geo} would be made infinitely small, this does not result in a sub-angstrom spot because the spherical aberration disc D_{Cs} and the diffraction disc D_{diffr} can not be made small simultaneously, considering their relationship with the convergence angle.
- D_{Cs} increases rapidly with increasing probe convergence angle. In contrast, the diffraction blur D_{diffr} gets smaller with increasing probe convergence angle. By the way, the chromatic aberration does not play a significant role here! It turns out that by selecting a decent small source image D_{geo} (with gunlens number = 6 and spot number = 9) and summing up all other contributions the smallest total probe diameter D_{tot} will be reached when setting the optimum probe convergence angle for Titan 300kV SuperTwin (without probe C_S corrector) to approximately 10 mrad when using the C2 aperture of 50 µm.

The strategy to set the probe convergence angle to the correct value is to first set it to 10 mrad directly using the flap-out Tune page of Titan's Beam Settings panel (see Figures 3), and then to use the Ronchigram for double checking and fine tuning of the probe convergence angle.

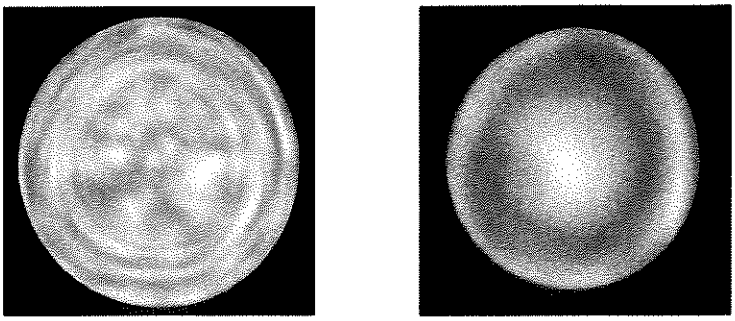


Figure 4: Ronchigrams (a) not at the best probe focus and (b) at the best focus with a right convergence angle.

A Ronchigram (Figure 4) is the shadow image of the probe illuminating the specimen, in diffraction mode. The procedure to observe a Ronchigram is:

- set specimen at eucentric height and focus image at eucentric plane (Eucentric Focus)
- select an amorphous area
- focus the probe to the sample with Intensity
- switch to diffraction
- fine tune probe focus (Intensity knob) until there are no details of the sample in the central part of the Ronchigram (termed "blow-up point")
- in case the Ronchigram is not circular symmetric, astigmatism needs to be corrected by using "Condenser stigmator"

The circular intensity variations in Figure 4a and Figure 4b are due to the spherical aberration. The dark rings at the centre of the Ronchigram now define visually how to tune the probe convergence angle for optimum HR-STEM resolution. The edge of the 50 µm C2 aperture should be at the bright ring just outside the first black ring around the central area, as is shown in Figure 4b. In case the beam limiting 50 µm C2 aperture does not cut-off the Ronchigram at the right position with a small difference, then the probe convergence angle needs to be optimize by changing the convergence angle in Tune page (Figure 3).

For reaching HR-STEM resolution it is required to have zero condenser two-fold astigmatism, and low three-fold condenser astigmatism. Both can be done by selecting the proper option in the Stigmator panel. Because of tolerances in the condenser system, the electron beam might pass off-axis through the Condenser C2 stigmator, resulting in a slight beam tilt inside the stigmator when doing stigmation. Therefore the "Rotation centre" of the objective-lens/condenser needs to be checked in Direct Alignments.

Here it is worth mentioning that beam tilt and beam shift need to be set correctly in Probe mode, too. To check the beam tilt, set the "Beam tilt pivot points" in Direct Alignments correctly with the specimen at the preset eucentric focus position in Probe mode. After confirming the correct C2 aperture position (as described above), the rotation centre needs to be carefully adjusted (probe concentrically in and out). This is the rotation centre of the upper + lower pole piece of the

objective lens. It will be discussed later, how to optimize the rotation centre further so that it is optimized on the upper pole piece of the objective lens only for an optimized probe.

The "Align Beam Shift" in "Direct Alignments" gives the possibility to make the beam jump to the centre of the screen after resetting the user beam shift. This needs to be done at a relatively high SA magnification.

3. (HR-)STEM mode

The specimen needs to be at the correct (eucentric) height. This is the imaging plane after pressing the "Eucentric focus" button. It needs to be checked if the objective lens current does not change when switching from Probe mode to STEM by looking at the objective lens (Obj %) reading on the Titan software display bar.

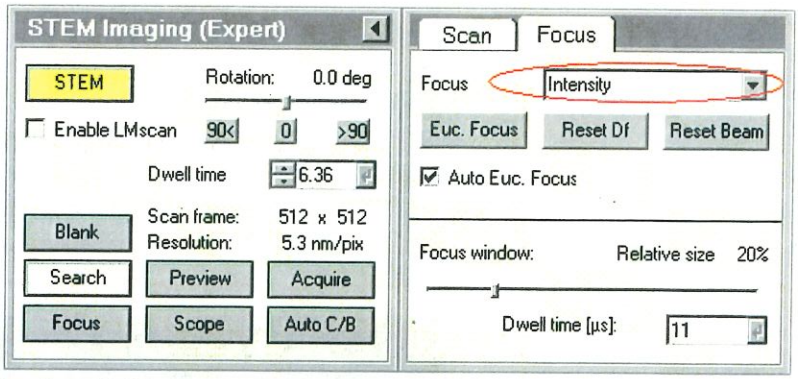


Figure 5: STEM Imaging control panel.

On a standard (non probe corrector) system, in STEM the focus knob is linked to the C3 lens, if the Focus selection is Intensity (figure 5). "Auto Euc. Focus" needs to be selected to make sure that the objective lens focus current in HM-STEM is at its preset eucentric value when switching to STEM. It is also very important to "Normalize all" the lenses, which can be set to a user defined button when switching the mode, and to press "Reset Beam" key in the Focus flap-out page (Figure 5) to check the beam shift.

For optimum HR-STEM imaging the rotation centre of the objective lens and the C2 aperture position need to be checked, and maybe be fine-tuned. The position of the C2 aperture determines, which part of the focusing condenser lens is *in use*. Ideally, the C2 aperture position is tuned such that the central (on-axis) part of the focusing condenser lens is used. The correct rotation centre setting makes sure that the beam (that exits on-axis the condenser lens) enters on-axis the objective lens. When the C2 aperture position is known to be correct, fine tuning the rotation centre can be done by using the "Direct Alignments". If the image of the C2 aperture in the Ronchigram is not centred on the symmetry centre then there is a mismatch between the aperture position and the rotation centre. After re-checking the position of the C2 aperture as has been described previously, the "Rotation centre" should be fine-tuned using the "Direct Alignments" procedure.

For optimal image contrast it is important to choose the right camera length and the correct HAADF detector settings. Using a standard Titan 300kV SuperTwin system, camera length of around 200mm is optimal for HAADF contrast (Z-contrast). Higher camera lengths would include Angular Dark Field contrast (the reverse of bright field contrast) into the STEM image. Brightness

level of the HAADF detector should be set to about 43%, and the Contrast level typically to values between 30% and 70% (Figure 6).

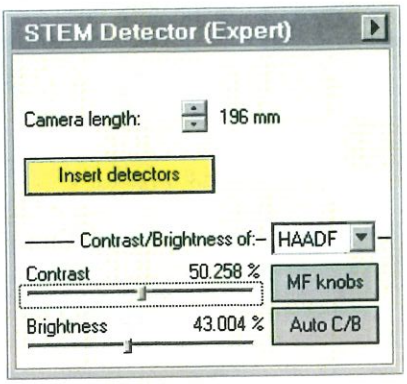


Figure 6: STEM Detector control panel.

4. HR-STEM imaging

The specimen typically used for HR-STEM proof-of-performance is the silicon specimen perfectly aligned in <110> zone axis. It shows (line) spacing of 3.1Å, 2.7Å, 1.9Å, 1.6 Å, and 1.36Å (dumbbell distance). These spacing are quite helpful in terms of fine-tuning focus and astigmatism, as well as specimen position and specimen tilt even before obtaining the final resolution.

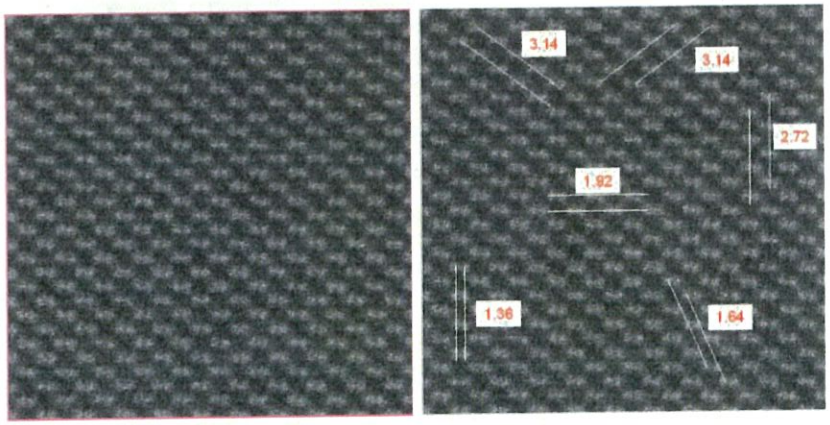


Figure 7: HR-STEM image of silicon <110> specimen

An area on the silicon sample which has the right quality (right thickness, thin/no amorphous layer, correct zone axis orientation) needs to be selected. Leave diffraction mode, normalize all lenses and focus the probe (intensity). Switch back to diffraction mode and move the sample in focus (correct z-height) with compustage-z ("blow up" point in Ronchigram).

The intensity ratio between bright field disc and scattered discs is a helpful criterion to determine if the specimen has the correct thickness. They should have approximately the same intensity. If the sample is too thin than the intensity of the bright field discs is much stronger than the scattered discs, therefore there will be no sufficient signal on the detector to resolve 1.36Å. If the sample is too thick the intensity of the bright field disc will be weak. Then electrons will scatter multiple times in the sample thus destroying resolution.

Important issues during HR-STEM sessions:

- Maintain specimen at the preset focus height. The optics performance has been optimized for this focus plane. This can be checked by switching from diffraction back to imaging, and inspecting the focused probe in relation to the sample.
- Regularly go back to the amorphous edge to check the quality of the Ronchigram. Pay attention to astigmatism and the convergence angle.

WORLD HEADQUARTERS
5350 NE DAWSON CREEK DRIVE
HILLSBORO, OREGON 97124 USA
PH: +1.503.726.7500

FEI EUROPE
PH: +31.40.23.56000

FEI JAPAN
PH: +81.3.3740.0970

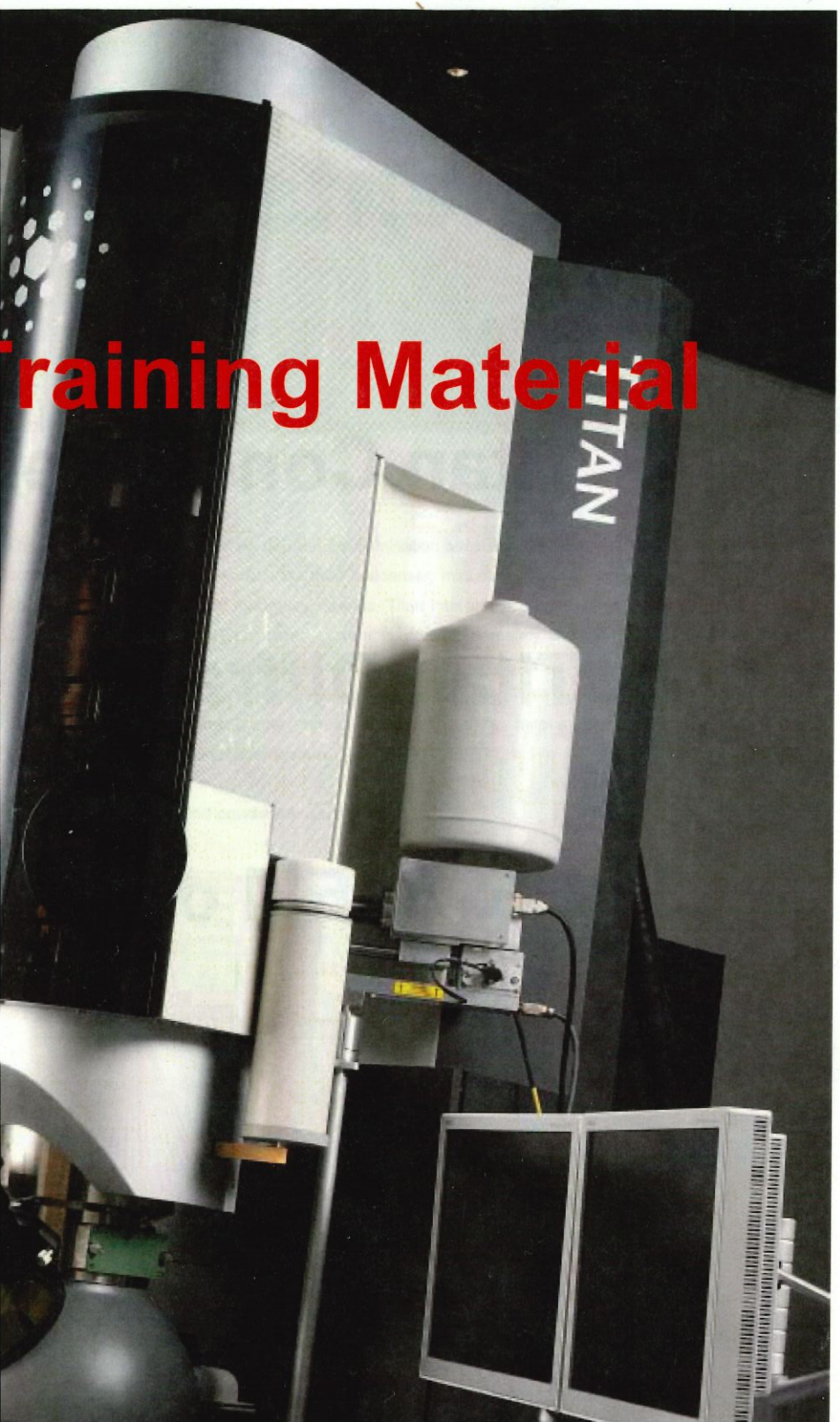
FEI ASIA PACIFIC
PH: +86.21.6122.5988

fei.com

TUV Certification for design, manufacture, installation and support of focused ion- and electron-beam microscopes for the NanoElectronics, NanoBiology, NanoResearch and Industry markets.

FEI COMPANY™
TOOLS FOR NANOTECH

© 2008. We are constantly improving the performance of our products, so all specifications are subject to change without notice. The FEI logo and Tools for Nanotech are trademarks of FEI Company.


raining Material
TITAN